

# Understanding Economic Effects of Natural Resource Extraction: The Importance of Linkages\*

Ann-Kristin Becker<sup>†</sup>

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## Abstract

This paper examines how the strength of local production linkages, that is, the extent to which activity in one sector stimulates other sectors through input demand and output supply, shapes the long-run impact of resource extraction on regional development. Exploiting a natural experiment in Germany (1849–2024), I compare the effects of extracting two types of coal—hard coal and lignite—that were mined under similar institutional conditions but differed in their degree of integration into local supply chains. Using an event-study framework and a shift-share design, I find that hard coal mining, which produced strong upstream and downstream linkages, led to persistent increases in local manufacturing and service employment, particularly in input-supplying and coal-using industries. In contrast, lignite mining, with weaker linkages, generated limited spillovers and contributed to industrial decline. I further show that the phase-out of hard coal after 1960 led to substantial employment losses, particularly in formerly linked sectors. These results highlight the double-edged nature of production linkages: they fuel growth during extraction but can deepen decline when extraction ends.

**Keywords:** Resource Extraction; Production Linkages; Local Multiplier; Structural Change; Deindustrialization

**JEL Codes:** J21, O14, Q32, Q33, R11

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<sup>†</sup>University of Cologne, Albertus-Magnus-Platz, 50923 Cologne, Germany; ([ann-kristin.becker@wiso.uni-koeln.de](mailto:ann-kristin.becker@wiso.uni-koeln.de))

# 1 Introduction

Natural resource extraction has long shaped regional development, from the coalfields of Europe to the oil towns of North America and the mining zones of Latin America and Sub-Saharan Africa. Yet its long-run local effects remain strikingly heterogeneous: some regions experience lasting industrial growth, while others face deindustrialization or persistent decline.<sup>1</sup> Despite extensive empirical and theoretical work, the mechanisms behind this divergence remain poorly understood. Why do some resource-rich regions thrive while others fall behind? This question is important both for regions that rely on resource extraction and for those transitioning away from it, since past specialization in extractive industries may limit their capacity to adapt to structural transformation.

In this paper, I investigate one potential mechanism behind the heterogeneity in the economic effects of natural resource extraction: the strength of local production linkages, that is, the extent to which activity in one sector stimulates economic activity in other sectors through input demand and output supply. I exploit a natural experiment in Germany (1849–2024), comparing the extraction of two types of coal—hard coal and lignite—that were mined under similar national institutions and at comparable scales, but differed markedly in their degree of local economic integration. Hard coal extraction required complex underground mining techniques and generated strong upstream linkages through demand for machinery, engineering, and metalworking. It also served as a key input in a wide range of manufacturing sectors, creating substantial downstream linkages. In contrast, lignite was extracted through simpler surface mining and used largely for electricity generation and local heating, resulting in much weaker ties to the surrounding industrial base. This setting enables a within-country comparison of two resource sectors that differ in linkage intensity but are otherwise similar in institutional context and scale, providing a credible framework for isolating the role of local economic integration in shaping the long-run regional effects of extraction. The gradual coal phase-out beginning in the 1960s further allows me to examine how regions with different linkage structures responded to a sustained decline in resource activity, a very relevant question for economies undergoing transitions away from fossil fuels today.

This paper builds on insights from staple theory, which emphasizes that the long-run impact of

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<sup>1</sup>While several studies present cases where natural resource extraction negatively impacts economic development (Esposito & Abramson, 2021; Papyrakis & Raveh, 2014; Shao et al., 2020), others demonstrate instances where resource extraction has positive local economic effects (Aragón & Rud, 2013; Cavalcanti et al., 2019; Michaels, 2011). For overviews of heterogeneous local effects see Havranek et al. (2016) and Ploeg (2011).

extracting a natural resource, or staple, depends on the extent to which it creates linkages to other sectors of the economy (Gunton, 2003; Watkins, 1963). I complement this perspective with a formal local multiplier framework adapted from Moretti (2010a) to derive testable predictions about how extractive industries shape regional labor markets. Specifically, I hypothesize that sectors like hard coal mining, which are embedded in local production networks through upstream demand for inputs and downstream use in industrial processes, generate stronger spillovers to other tradable sectors and amplify demand for local services. By contrast, sectors that operate as enclaves, such as lignite mining, with limited integration into local supply chains, are less likely to produce such spillovers and may even crowd out other tradables by raising input costs.<sup>2</sup> Importantly, the same linkages that support growth during extraction may increase exposure to decline. I therefore expect hard coal regions to experience larger gains during the extraction period but also sharper contractions during the coal phase-out. Lignite regions, by contrast, are likely to see limited benefits, or even adverse effects, during extraction, but smaller losses during decline. I test these hypotheses empirically by comparing the long-run effects of hard coal and lignite extraction in Germany. Since both resources were mined under similar national institutions and at comparable scales, but differed markedly in their degree of local economic integration, this setting allows me to isolate the role of production linkages in shaping regional development.

To examine whether differences in local linkage intensity shaped the long-term effects of resource extraction, I construct a novel county-level panel dataset for Germany spanning 1849 to 2024, covering the rise, peak, and gradual decline of both the hard coal and lignite mining industry. I link this dataset to a historical input–output table to classify manufacturing sectors by their upstream and downstream linkages to coal mining, enabling a disaggregated analysis of sectoral spillovers.

An empirical challenge is that coal production may be endogenous to local economic conditions. To address this, I proxy for mining activity using spatial variation in geological endowments of hard coal and lignite across German regions. Since these deposits were formed millions of years before modern economic activity, their geographic distribution is plausibly exogenous.

To identify the local effects of coal extraction, I combine two complementary empirical strategies.

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<sup>2</sup>Higher mining employment can raise local wages, increasing production costs for all firms. Since tradable goods are sold in national or international markets, local producers cannot pass these costs on to consumers and may become uncompetitive.

First, I implement an event-study design comparing changes in employment across hard coal, lignite, and non-coal regions following the onset of large-scale extraction. This approach allows me to trace both the long-term effects of extraction and the impacts of the coal phase-out. Second, I estimate a shift-share specification that interacts annual variation in global coal prices with county-level exposure to coal deposits. This design allows me to estimate the cyclical effects of extraction on regional labor markets.

My findings show that the strength of local production linkages plays a central role in shaping the long-term effects of resource extraction. The event-study results show that hard coal mining, characterized by strong local linkages, led to sustained gains in manufacturing employment, with average increases of approximately 19% relative to non-coal regions. These gains were driven by employment growth in linked industries, those supplying inputs to mining (upstream), those using coal as an input (downstream), and those with both types of linkages. Service employment also increased, consistent with local demand spillovers from rising income and population, while agricultural employment declined as regions industrialized. In contrast, lignite mining, marked by weaker local linkages, was associated with industrial decline. Manufacturing employment fell by approximately 20% alongside similar contractions in the service sector, suggesting limited spillovers and crowding out of both tradable and non-tradable activities. The shift-share estimates broadly support the event-study findings. While the estimated magnitudes are smaller, reflecting that the shift-share design captures short-run responses to global price fluctuations, whereas the event study traces cumulative long-run effects, they reinforce the direction of effects using a distinct identification strategy. Following the phase-out of hard coal beginning in the 1960s, hard coal regions experienced pronounced deindustrialization, with manufacturing employment declining by about 31% relative to non-coal areas. Importantly, the vulnerability of hard coal regions was not due to low industrial diversification: I show that hard coal regions actually became more diversified during the extraction period. Instead, my results show that it was the depth of integration with the coal economy, particularly in sectors tightly linked to mining, that shaped the severity of losses once coal was phased out. Taken together, the results suggest that regions with strongly integrated extractive sectors tend to gain more during periods of extraction but face sharper contractions during phase-out. In contrast, regions where extractive industries are weakly linked to other local sectors may experience adverse effects during both the extraction phase and the subsequent decline,

although the losses during phase-out tend to be more limited.

**Contribution to the literature.** This paper contributes to four strands of research on the local economic effects of natural resource extraction by highlighting the role of production linkages in shaping regional outcomes. **First**, it adds to work on the heterogeneity of extraction impacts by showing that the direction and magnitude of local effects depend on the degree of economic integration between the extractive sector and the surrounding economy (Aragón & Rud, 2013; Black et al., 2005; Dube & Vargas, 2013; Lippert, 2014; Pelzl & Poelhekke, 2021). **Second**, it extends the literature on local labor markets by demonstrating that both the size and direction of spillovers to tradable and non-tradable sectors can vary substantially, even within a single industry, depending on linkage intensity (Greenstone et al., 2010; Moretti, 2010a, 2010b). **Third**, it contributes to research on structural change and regional resilience by showing that even relatively diversified regional economies can become highly vulnerable when their lead sectors are tightly embedded in local supply chains (Berbée et al., 2024; Black et al., 2005; Esposito & Abramson, 2021; Franck & Galor, 2021; Fritzsche & Wolf, 2022; Gagliardi et al., 2023; Glaeser et al., 2015; Jacobsen & Parker, 2016). Once such a lead sector contracts or disappears, the resulting disruption can amplify employment losses and trigger broader industrial decline. **Finally**, the findings speak to the broader resource-curse debate by showing that variation in local linkage structures helps explain why the effects of resource wealth differ so markedly across regions (Allcott & Keniston, 2018; Caselli & Michaels, 2013; Cust & Poelhekke, 2015; Havranek et al., 2016; Michaels, 2011; Ploeg, 2011; Sachs & Warner, 2001).

Taken together, the paper offers a unified explanation for why similar resource endowments can lead to sharply divergent development paths, depending on the strength of local production linkages. It shows that extraction generates lasting local benefits only when it creates strong production linkages to the broader economy. Where extraction remains isolated, regions tend to stagnate or decline. This mechanism helps reconcile conflicting findings in the resource literature and sheds light on why the legacy of resource wealth can range from long-run prosperity to persistent underdevelopment.

**Structure of the paper.** The remainder of the paper is organized as follows. Section 2 provides background on Germany’s coal mining industries, including evidence on differences in local supply chain integration between the hard coal and lignite sectors. Section 3 develops the

conceptual framework and derives testable hypotheses, drawing on the local multiplier model of Moretti (2010b). Section 4 describes the data, and Section 5 outlines the empirical strategy and identification approach. Section 6 presents the main findings, focusing on the long-run economic impacts of coal mining, the cyclical effects of resource booms and busts, and the mechanisms driving these outcomes. Section 7 assesses the robustness of the results. Section 8 discusses the implications and concludes.

## 2 Setting: Hard Coal and Lignite Mining in Germany

Germany provides a compelling setting to examine how variation in production linkages shapes the long-run local effects of resource extraction. For more than a century, the country extracted two major coal types: hard coal (anthracite and bituminous) and lignite (brown coal), both produced under a unified institutional and regulatory framework. Although the two sectors operated at comparable national scales and during overlapping periods, they differed substantially in their degree of integration into the surrounding economy. Hard coal was closely linked to upstream supplier industries and downstream industrial users, while lignite remained more economically isolated.

This contrast allows for a within-country comparison of resource sectors that differ in linkage intensity but are otherwise similar in scale and institutional context, providing a credible setting to isolate the role of production linkages in shaping long-run regional development, during both the expansion and decline of extractive activity.

### 2.1 Evolution of Coal Mining in Germany

While coal extraction in Germany began in the 18th century, it remained small in scale until the mid-19th century, when the diffusion of coal-based technologies from Britain both stimulated industrial demand and facilitated mining expansion (Tilly & Kopsidis, 2020).<sup>3</sup> Coal soon became the dominant energy source for Germany’s industrial economy. By the early 20th century, Germany had become one of Europe’s largest hard coal producers, with output levels comparable to Britain.

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<sup>3</sup>Between 1850 and 1900, Germany’s hard coal production increased more than 20-fold, while lignite production rose more than 26-fold (see Figure A1a). Over the same period, employment in hard coal mining rose more than 12-fold, and lignite employment increased over 8-fold (own calculations based on data from Fischer and Fehrenbach (1989)).

Germany also emerged as the leading producer of lignite in Europe (Fremdling, 1995a). While hard coal production initially dominated, lignite output caught up during the interwar period. From the 1920s onward, hard coal and lignite were produced at similar scales (see Figure A1).<sup>4</sup> By the 1960s, German hard coal production declined strongly. High labor costs, competition from cheaper imported coal, and the growing adoption of alternative energy sources, such as oil, natural gas, and nuclear power, undermined its competitiveness. Between 1960 and 1985, hard coal production fell by nearly 50 percent, roughly 85 percent of mines closed, and employment in the sector declined by more than two-thirds (own calculations based on Fischer and Fehrenbach (1989)). The last hard coal mine in Germany closed in 2018. Lignite remained economically viable until the early 1990s, as technological improvements in surface mining, delayed its decline. By the 1990s, lignite output began to decline as stricter environmental regulations were introduced. Between 1990 and 2023, lignite output declined by more than two-thirds (own calculation based on (Statista Research Department, n.d.-a)). Although lignite extraction remains active today (as of 2025), it is scheduled to be fully phased out by 2035 as part of Germany’s broader energy transition. This long historical trajectory enables analysis of both the short- and long-run effects of extractive activity and its gradual decline.

## 2.2 Institutional Context

Hard coal and lignite were extracted under broadly similar institutional and regulatory frameworks. Both sectors were governed by the Prussian Mining Law of 1865 and its subsequent adaptations, which standardized licensing, safety regulations, and labor relations across regions. Mining was primarily carried out by private firms, and while municipalities collected some business taxes (Gewerbesteuer), most mining-related revenues accrued to state or national governments. There were no major differences in regulatory treatment or subsidy regimes between hard coal and lignite (Fremdling, 1995b; Tilly & Kopsidis, 2020). As a result, persistent regional differences are more likely to reflect variation in production linkages than in institutions or governance.

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<sup>4</sup>This catch-up was enabled by key technological innovations, most notably Exter’s lignite briquetting press, which improved the energy density and transportability of lignite.

## 2.3 Geographic Distribution of German Coal Resources

Coal extraction in Germany was geographically concentrated in several distinct regions, with the two coal types—hard coal and lignite—mined in largely non-overlapping parts of the country. Major hard coal deposits were located in central and western Germany, particularly in the Ruhr region (e.g., the districts of Arnsberg and Düsseldorf), the Saar area (e.g., Trier), and in Upper Silesia (e.g. Oppeln, now part of modern Poland). In contrast, lignite was mined mainly in the Rhenish mining district (e.g., Köln), the Lusatian district (e.g., Frankfurt-Oder), and the Central German mining district (e.g., Merseburg) (Kunz, 2008). Figure 3 illustrates the distribution of coal fields in Germany, which closely corresponds to the historical location of coal mines (see also Section 4.1). The clear geographic separation between hard coal and lignite deposits, combined with their broad dispersion across Germany, provides the variation needed to credibly compare local economic outcomes across regions exposed to hard coal, lignite, or no coal mining, while holding institutions constant.

## 2.4 Differences in Production Linkages Between Hard Coal and Lignite

Hard coal and lignite mining differed markedly in their potential to generate local economic spillovers through input–output linkages. I distinguish between two types: upstream linkages, reflecting the mining sector’s demand for local intermediate inputs, and downstream linkages, reflecting its role as a supplier to other local industries.

### 2.4.1 Upstream Linkages: Coal Mining’s Dependence on Local Inputs

Hard coal mining relied more heavily on upstream industries than lignite mining, reflecting its more complex extraction process. Because hard coal lies deep underground, its extraction required shafts, tunnels, drainage systems, and specialized equipment, all of which demanded advanced machinery and industrial materials. By contrast, lignite is typically located close to the surface and extracted using open-pit mining, which involves removing topsoil to access shallow seams. This method required fewer specialized inputs and less advanced technology (Hovis & Mouat, 1996). As a result, hard coal mining generated stronger upstream linkages to supplier industries such as machinery, timber processing, and metal fabrication, whereas lignite extraction was less dependent on external

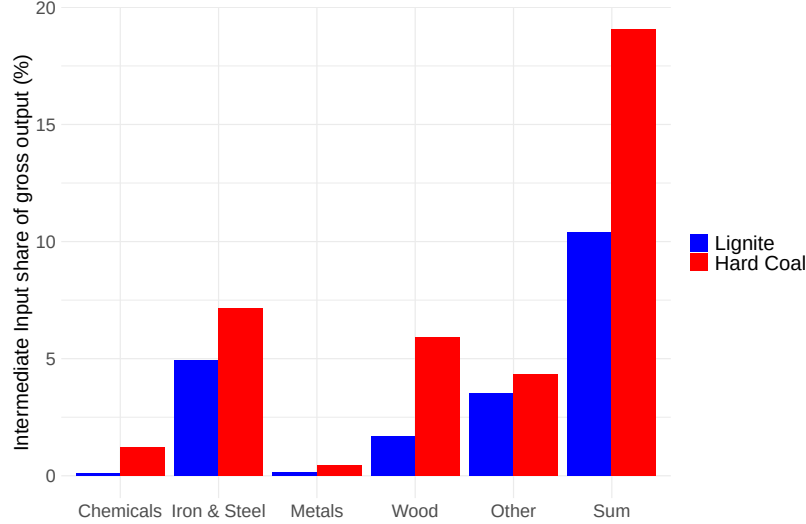


Figure 1: Hard Coal and Lignite Mining Intermediate Inputs in Germany (1936)

*Notes:* The figure plots the value of inputs purchased from various sectors by the lignite (blue) and hard coal (red) mining industries in Germany in 1936, expressed as a share of each sector’s total output. Own calculations based on data from the Statistisches Reichsamt (1936a).

supplier industries.

Figure 1 illustrates this difference. In 1934, intermediate inputs accounted for 19.1% of total production value in hard coal mining, compared to just 10.4% in lignite. Moreover, specialized materials such as mine timber and explosives made up 6.3% of hard coal output but only 1.5% in lignite (Statistisches Reichsamt (1936a), own calculations).

Where institutional and economic conditions permit, there are strong incentives to source mining inputs locally: proximity reduces transportation costs and delivery risks, facilitates customized responses to operational needs, and can foster innovation through firm-level collaboration (Morris et al., 2012). As a result, hard coal extraction’s greater reliance on complex, locally sourced inputs fostered the growth of supplier industries and generated stronger upstream industrial spillovers.

#### 2.4.2 Downstream Linkages: Local Use of Coal

Hard coal can be used across a broader range of industries than lignite, particularly in energy-intensive processes such as steel and iron production, resulting in stronger downstream linkages. Lignite has a lower energy density due to its higher moisture content and lower carbon composition, producing less energy per unit of mass when burned. In addition, lignite’s chemical properties

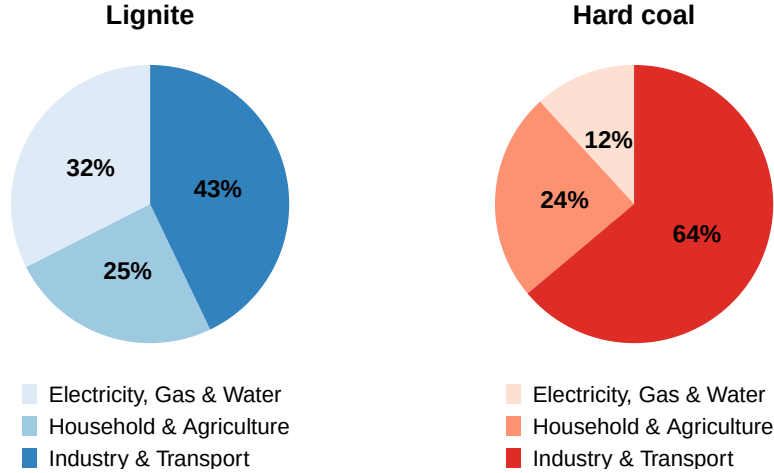


Figure 2: Hard Coal and Lignite Use in Germany (1934)

*Notes:* The figure shows the sectoral shares of lignite (left) and hard coal (right) use in Germany in 1934. Own calculations based on data from the Statistisches Reichsamt (1934).

prevent its use in coke production, an essential input for steelmaking, heavy industries, and blast furnaces in metallurgical processes.

These differences in energy content and usability translated into distinct patterns of industrial coal use, as shown in Figure 2. In 1934, lignite consumption was more heavily concentrated in electricity generation and small-scale uses such as household heating and agriculture, with relatively smaller use in industrial production compared to hard coal. While these uses generated localized demand, they were less likely to trigger large-scale industrial spillovers. By contrast, hard coal was primarily used in industrial applications, particularly in energy-intensive sectors such as heavy manufacturing, metal processing, and transport.

Processing coal near extraction sites offered clear economic advantages: it was bulky, degraded quickly, and was costly to transport and store. These constraints encouraged the co-location of downstream industries in hard coal regions. Consequently, lignite remained primarily confined to non-industrial uses, while hard coal was employed across a broader range of sectors, including energy-intensive manufacturing and transportation. As a result, downstream spillovers, via industrial demand and spatial co-location, were likely more pronounced in hard coal regions.

Taken together, the sharp contrast in linkage intensity between hard coal and lignite, across otherwise comparable sectors, offers a rare opportunity to isolate how differences in input–output structures shape the long-run local effects of resource extraction.

### 3 Conceptual Framework: Sectoral Spillovers from Coal Mining

To study how coal mining affects employment across sectors depending on linkage intensity, I develop a simple general equilibrium framework adapted from Moretti (2010b). I model each county as a competitive economy that produces tradable goods (mining, manufacturing, agriculture) sold on national markets and non-tradable goods (services) whose prices are set locally.<sup>5</sup> Labor is homogeneous and mobile across sectors within a county, implying that wages equalize locally across industries. The local labor supply curve is upward sloping: workers' utility depends on local wages net of living costs, along with idiosyncratic location preferences. Housing supply is also upward sloping, reflecting geographic constraints and land-use regulation. Amenities are assumed to be constant across space. This framework allows me to derive testable predictions about how extractive sectors with varying linkage intensities shape local labor markets.

**Spillovers to the Tradable Sector.** An increase in mining employment raises local wages, which in turn increases production costs for all local firms. Since tradable goods are sold in national or international markets, local producers cannot pass these higher costs on to consumers (they are price-taker), potentially leading to crowding-out effects. However, if mining generates upstream or downstream linkages with other local industries, it may stimulate activity in related sectors. The net impact depends on the strength of these production linkages and the presence of agglomeration economies, such as productivity gains from industry clustering.

In the case of hard coal, both upstream and downstream linkages are strong, as outlined in Section 2.4. Mining depends on industrial inputs and simultaneously supplies fuel to a broad range of industries. These linkages are likely to generate positive spillovers to nearby manufacturing firms, which may expand employment in response to increased input demand from mining or improved access to coal as an input. By contrast, agriculture, lacking direct connections to mining, primarily faces cost-side pressures: rising wages and land prices reduce profitability, potentially leading to employment declines.

**Hypothesis H1:** *Manufacturing employment increases in hard coal regions, particularly in sectors with strong input–output linkages.*

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<sup>5</sup>This distinction follows Moretti (2010a), based on the geographic scope of product markets.

**Hypothesis H2:** *Agricultural employment declines in hard coal regions due to rising input costs and land competition.*

In lignite regions, production linkages are relatively weak. Lignite is primarily used for local heating and electricity generation, and its extraction process relies on fewer intermediate inputs. As a result, manufacturing firms benefit little from proximity to mining activity but may still face upward wage pressure, potentially reducing employment. The effects on agriculture are more ambiguous. On the one hand, rising wages may increase production costs; on the other hand, if mining displaces other sectors and weakens overall labor demand, wage pressures may ease over time, potentially stabilizing agricultural employment.

**Hypothesis L1:** *Manufacturing employment in lignite regions declines due to weak linkages and rising wages.*

**Hypothesis L2:** *Agricultural employment in lignite regions remains stable or increase if wage pressures are offset by reduced demand in other sectors.*

**Spillovers to the Non-Tradable Sector.** Higher mining wages and employment raise local incomes, which in turn increase demand for non-tradable goods and services. The magnitude of this demand linkage effect depends on the labor intensity of services and the share of income spent locally. In addition, mining operations may generate direct demand for services such as logistics, maintenance, and catering.<sup>6</sup>

In hard coal regions, both income-driven and mining-related service demand are expected to raise employment in the non-tradable sector. In lignite regions, by contrast, weaker income gains and limited service linkages may result in stagnant or declining service employment, particularly if population falls due to out-migration or crowding-out of other sectors.

**Hypothesis H3:** *Service employment increases in hard coal regions due to higher income and mining-related demand.*

**Hypothesis L3:** *Service employment stagnates or falls in lignite regions due to limited income growth and weak service demand from mining.*

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<sup>6</sup>Moretti (2010b) estimates that each additional manufacturing job generates approximately 1.6 jobs in local services.

**Total Employment Effects.** In regions with strong input–output linkages, mining acts as a catalyst for broader economic expansion. Hard coal regions should experience net employment growth across sectors, despite agricultural losses. In lignite regions, weak spillovers and higher costs may limit overall gains or lead to net employment losses.

**Hypothesis H4:** *Total employment increases in hard coal regions due to strong local multipliers.*

**Hypothesis L4:** *Total employment effects in lignite regions are modest or negative due to weak spillovers and crowding-out effects.*

**Effects of Coal Phase-Outs.** The framework also applies to gradual declines in mining activity, such as those associated with long-term coal phase-outs. These contractions reverse the mechanisms described above.

In hard coal regions, strong upstream and downstream linkages imply that declining mining employment reduces demand in manufacturing and services, triggering broader employment losses. In lignite regions, by contrast, impacts are more concentrated within mining itself. Still, falling incomes and population may reduce demand for local services, leading to additional employment losses in the non-tradable sector.

**Hypothesis H5:** *A decline in hard coal mining reduces employment in manufacturing and services due to strong production and demand linkages. Effects on agriculture are ambiguous and may depend on how wage and land cost pressures adjust.*

**Hypothesis H6:** *In lignite regions, employment losses are concentrated in services, while manufacturing and agriculture are less affected.*

I test these predictions using two complementary empirical strategies. First, I employ an event-study framework to examine the long-run effects of mining and the coal phase-out. Second, I implement a shift–share design that interacts global coal price shocks with county-level exposure to coal endowments, capturing the dynamic effects of resource booms. Together, these strategies allow me to isolate the causal impact of coal extraction and distinguish between short-run boom–bust cycles and persistent long-term effects.

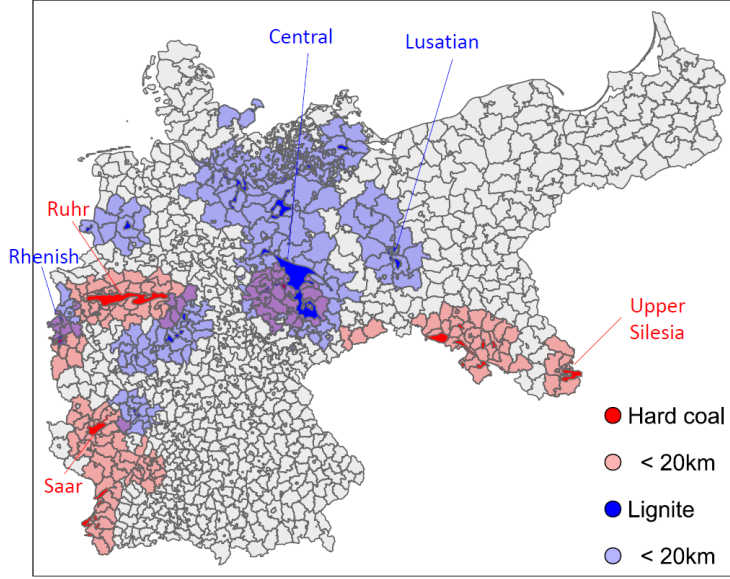


Figure 3: Geographic Locations of Hard Coal and Lignite Deposits in Germany

*Notes:* The map displays hard coal (red) and lignite (blue) deposits within the borders of Germany as of 1900. Counties classified as hard coal or lignite regions in the analysis, i.e. those located within 20km of the respective deposits, are highlighted accordingly. Major coal-producing areas (e.g., Ruhr, Saar, Upper Silesia) and lignite basins (e.g., Rhenish, Central German, Lusatian) are labeled. Data source: Asch and Bellenberg (2005).

## 4 Main Data

This section describes the construction of the main dataset used in the analysis. The unit of observation is the county. To ensure spatial consistency over time, I harmonize administrative boundaries to match those in place in 1900 (see Appendix A.4).

### 4.1 Treatment Variables: Hard Coal and Lignite Endowments

To measure exposure to coal mining, I use geological data on the location of hard coal and lignite deposits across Germany. Rather than relying on historical mining activity, which may be endogenous to prior economic development, I follow Fernihough and O'Rourke (2021) in using geological resource endowments, based on data from Asch and Bellenberg (2005), as a plausibly exogenous proxy for mining potential.

I classify strata from the Carboniferous period (ca. 320–295 million years ago) as hard coal-bearing, since nearly all German hard coal was formed during this era. Lignite, by contrast, is typically found in younger formations dating from the Miocene to Pliocene periods (ca. 23–3 million

years ago). Because these younger formations are geologically more heterogeneous, I restrict lignite classification to areas where lignite is among the two most prevalent rock types. I then compute the distance from each county’s geographic centroid to the nearest hard coal or lignite-bearing strata. As shown in Figure A2a and Figure A2b, proximity to hard coal and lignite formations aligns closely with the location of historical coal fields. I classify a county as a hard coal (or lignite) region if its centroid lies within 20 kilometers of the corresponding strata. Alternative thresholds are tested in robustness checks (see Section 7.1).

Table A1 presents summary statistics. On average, counties are located 132 kilometers from the nearest hard coal deposit and 110 kilometers from the nearest lignite deposit. Based on the 20 kilometers cutoff, 16% of counties are classified as hard coal regions and 18% as lignite regions. About 30% of counties are located near either type of deposit, with roughly 3% lying near both.

## 4.2 Outcome Variables: County-Level Sectoral Employment

To analyze the local effects of coal resource endowments, I compile a novel dataset of county-level employment spanning 1849 to 2024, drawn from historical occupational censuses. The data capture the rise and expansion of Germany’s hard coal and lignite industries, as well as the onset of the coal phase-outs beginning in the 1960s.<sup>7</sup> This long-run panel allows me to trace the economic effects of coal endowments during both the extraction phase and the gradual decline of coal mining.

### 4.2.1 Disaggregated Manufacturing Employment and Production Linkages

To examine heterogeneity in the effects of coal mining across manufacturing industries, I construct measures of upstream and downstream linkages between each industry and the mining sector. These linkage intensities are derived from the 1947 U.S. Input-Output (I-O) table compiled by the U.S. Bureau of Economic Analysis (2019), which provides detailed information on inter-industry transaction flows.<sup>8</sup>

For each manufacturing industry, I calculate first- and second-degree upstream and downstream linkage shares. The first-degree downstream linkage is defined as the share of an industry’s total

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<sup>7</sup>Sectoral employment data are harmonized across time and adjusted for administrative boundary changes to construct a consistent panel at the 1900 county level (see Section A.4). Section A.5 describes the underlying sources.

<sup>8</sup>I rely on the U.S. I-O table to avoid potential endogeneity concerns associated with using historical German tables, which may themselves have been shaped by coal mining activity. In robustness checks, I confirm that the results are not sensitive to the choice of I-O table by replicating the analysis using a historical German benchmark.

inputs sourced directly from the mining sector:<sup>9</sup>

$$\text{FirstDegreeDownstream}_i = \frac{\text{InputFromMining}_i}{\text{TotalInput}_i}$$

The first-degree upstream linkage is defined as the share of an industry’s output sold directly to the mining sector:

$$\text{FirstDegreeUpstream}_i = \frac{\text{OutputToMining}_i}{\text{TotalOutput}_i}$$

Second-degree linkages capture indirect relationships via intermediate industries and are calculated analogously through two-step I-O matrix multiplications. An industry is classified as upstream-linked if its total upstream linkage (direct plus indirect) exceeds the sector median. To prevent electricity-related sectors from dominating the downstream classification, I follow Allcott and Keniston (2018) and use only the direct input share. Industries are considered downstream-linked if their direct input share from mining exceeds the median. An industry is defined as linked to mining, if it is classified as either upstream- or downstream-linked; otherwise, it is considered non-linked.

Appendix Table A2 reports the linkage shares and binary indicators for each manufacturing industry. The strongest upstream linkages are found in *Machinery* and *Wood products*, both of which supply key inputs to mining operations. Industries with the highest downstream linkages are *Nonmetallic mineral products* and *Primary metals*, reflecting their reliance on coal as an energy or material input. Finally, I map census occupations for each available year to the I-O sectors, allowing disaggregated analysis of sector-specific employment dynamics.<sup>10</sup>

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<sup>9</sup>The mining sector itself is excluded from linkage calculations.

<sup>10</sup>Disaggregated manufacturing employment is available for the years 1849, 1882, 1895, 1907, 1939, 2007, 2015, and 2024.

## 5 Empirical strategy

### 5.1 Event-study: Long-Run Effects of Resource Extraction and Its Phase-Out

To assess the long-run effects of hard coal and lignite mining on local economies and to evaluate the effects of the coal phase-out, I estimate the following event-study specification:

$$Y_{ct} = \sum_{j=t_1}^T \beta_j (HardCoal_c \times I_t^j) + \sum_{j=t_1}^T \delta_j (Lignite_c \times I_t^j) + \sum_{j=t_1}^T (X_c' I_t^j) \kappa_j + \alpha_c + \gamma_t + \varepsilon_{ct}. \quad (1)$$

Here,  $Y_{ct}$  is the outcome of interest in county  $c$  and year  $t$ . The indicators  $HardCoal_c$  and  $Lignite_c$  equal 1 if the county centroid lies within 20 kilometers of a hard coal or lignite deposit, respectively. These indicators are interacted with year dummies  $I_t^j$  to estimate time-varying treatment effects. The coefficients  $\beta_j$  and  $\delta_j$  capture the difference in outcomes between hard coal and lignite counties, respectively, in year  $j$ , relative to a baseline year.<sup>11</sup>

The vector  $X_c$  includes time-invariant geographic controls (temperature, rainfall, slope, crop suitability, distance to the nearest river and coast) interacted with year fixed effects to flexibly account for time-varying confounders. County fixed effects  $\alpha_c$  absorb all time-invariant unobserved heterogeneity, while year fixed effects  $\gamma_t$  control for common shocks affecting all counties. Standard errors are clustered at the county level.

Identification relies on the standard parallel trends assumption: in the absence of coal endowments, counties with and without deposits would have followed similar trends in the outcome variable, conditional on controls. This assumption is plausible given that coal deposits were formed by natural processes millions of years before modern economic activity. I further assess its validity by examining pre-trends in an alternative outcome (wages) in Section 7.7.

### 5.2 Reduced form Shift-Share Design: Effects of Coal Booms

To complement the event-study analysis and examine how local outcomes respond to global coal market fluctuations, I estimate a reduced-form shift-share specification in which geological endow-

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<sup>11</sup>Due to changes in territorial borders, I estimate the model separately for (i) the period of coal industry expansion and (ii) the era of coal phase-out, using distinct time windows and baseline years for each subsample.

ments provide the exposure shares, and global coal prices serve as the time-varying shocks:

$$Y_{ct} = \beta_H \text{HardCoal}_c \times \text{WorldPrice}_t + \beta_L \text{Lignite}_c \times \text{WorldPrice}_t + \sum_{j=t_1}^T X'_c I_t^j \delta_j + \alpha_c + \gamma_t + \epsilon_{ct}. \quad (2)$$

Here,  $\text{HardCoal}_c$  and  $\text{Lignite}_c$  are binary indicators equal to 1 if county  $c$  lies within 20 kilometers of a hard coal or lignite deposit, respectively. The shift variable  $\text{WorldPrice}_t$  captures fluctuations in global coal prices and is standardized to have mean zero and standard deviation one. Coefficients  $\beta_H$  and  $\beta_L$  thus capture the differential effect of a one-standard-deviation increase in global prices on outcomes in hard coal and lignite regions, respectively, relative to non-endowed counties. All specifications include county fixed effects ( $\alpha_c$ ), year fixed effects ( $\gamma_t$ ), and a vector of time-varying geographic controls ( $X'_c I_t^j$ ).

This design complements the event-study framework by leveraging a distinct source of identifying variation. While the event-study rests on the parallel trends assumption, the shift-share approach relies on the interaction between time-invariant exposure and time-varying shocks, and can identify causal effects under two alternative assumptions:

- Share exogeneity: Conditional on controls, the geological exposure shares ( $\text{HardCoal}_c$ ,  $\text{Lignite}_c$ ) are uncorrelated with unobserved county-specific shocks (Goldsmith-Pinkham et al., 2020).
- Shift exogeneity: Conditional on controls, global coal price movements are exogenous to local German economic conditions (Adao et al., 2019; Borusyak et al., 2022).

Both assumptions are plausible in this context. The location of coal deposits was determined by ancient geological processes, not by modern economic activity. Similarly, global coal prices were driven primarily by international markets and are unlikely to be influenced by German economic conditions. The shift-share design also offers several practical advantages. By leveraging global coal booms and busts, I can begin the analysis in 1882, which allows me to include all counties within the historical borders of the German Empire. Additionally, price shocks provide more precise timing of economic cycles than the gradual and continuous treatment used in the event-study. While local labor market responses may not be immediate, due to adjustment frictions,

the shift–share design helps distinguish short-run boom–bust dynamics from longer-run structural effects of resource extraction and reinforces identification through an alternative source of variation.

## 6 Main Results

### 6.1 Long-Run Effects of Resource Extraction: Event-study Results

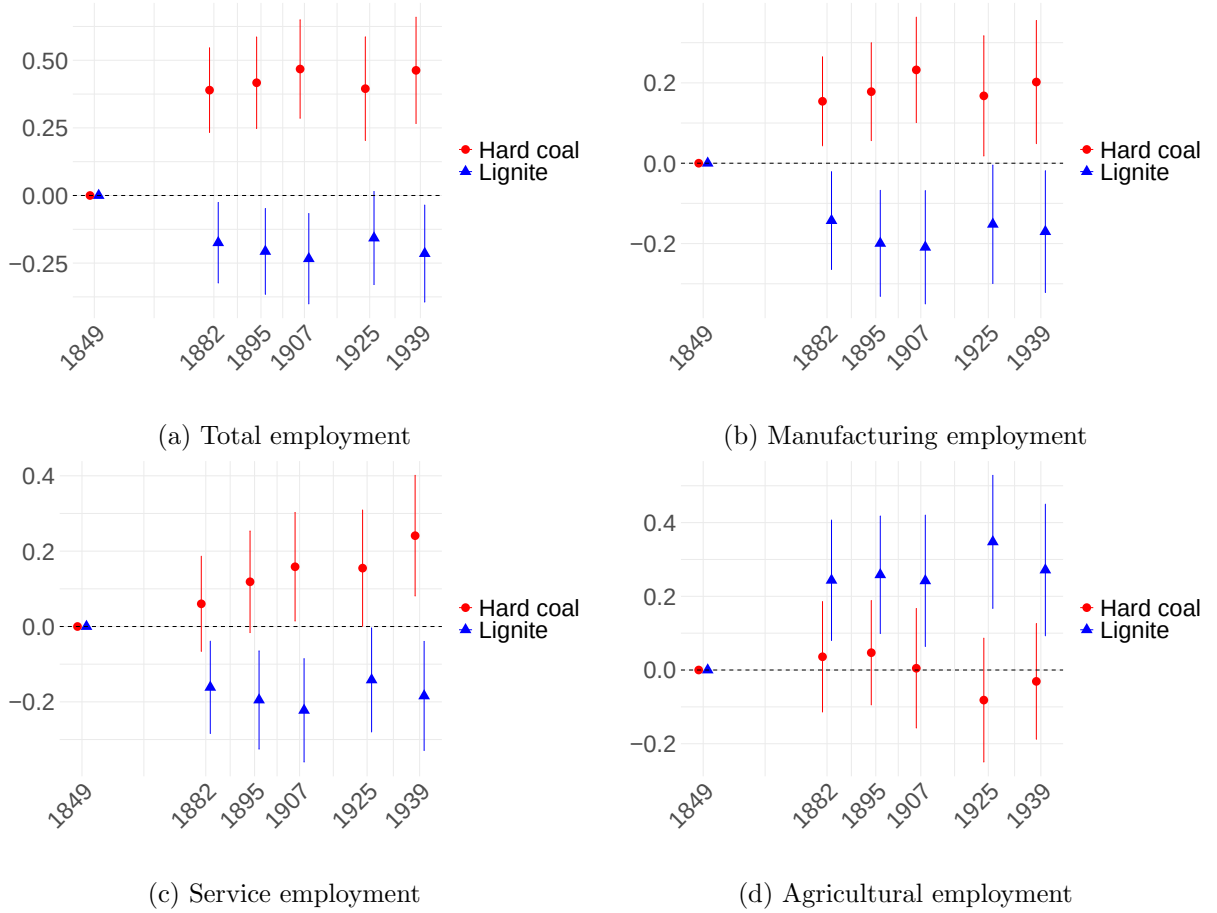


Figure 4: Event-Study: Effect of Resource Endowment on Sectoral Employment (1849–1939)

*Note:* This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a standard deviation of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties. Corresponding estimates for each plot are presented in Table A4 in the Appendix.

I begin by examining the long-run labor market effects of large-scale coal extraction. I use

1849 as the pre-treatment baseline, as coal production was limited to small-scale operations prior to this point. Industrial-scale mining expanded in the decades that followed, with employment effects expected to materialize by the first treatment year, 1882 (see Figure A1).<sup>12</sup> The event-study estimates thus capture employment changes following the onset of large-scale coal extraction.

Figure 4 shows that the local effects of coal extraction varied sharply by coal type, consistent with the role of production linkages outlined in Section 3. Total employment increased by 0.3 to 0.5 (log) standard deviations between 1849 and 1882, and remained persistently elevated through 1939 (Panel 4a).<sup>13</sup> In contrast, lignite regions saw a sustained decline of 0.1 to 0.2 (log) standard deviations below baseline, consistent with limited economic spillovers and potential crowding-out of other sectors. These patterns align with Hypothesis H4, which anticipates broad employment gains in hard coal regions due to strong production linkages and local multipliers, and with Hypothesis L4, which predicts limited or negative effects of lignite extraction due to weak intersectoral connections. In manufacturing (Panel A4b), employment in hard coal regions rose by 0.2 to 0.3 (log) standard deviations, in line with strong upstream and downstream linkages (Hypothesis H1). Lignite regions saw declines of similar magnitude, reflecting weak intersectoral connections and crowding-out effects (Hypothesis L1). A similar pattern emerges in the service sector (Panel 4c): employment in hard coal regions rose, likely reflecting demand spillovers from rising population, income, and mining-related service needs (Hypothesis H3). Service employment in lignite areas declined over time, likely reflecting a shrinking workforce, reduced local demand, and limited demand-side linkages from mining activity (Hypothesis L3). In agriculture (Panel 4d), hard coal regions show no significant change, consistent with offsetting forces from wage pressures and land competition (Hypothesis H2). In contrast, agricultural employment in lignite regions increases modestly, likely reflecting reallocation into lower-productivity sectors as manufacturing and services contracted (Hypothesis L2).

To assess the magnitude of these effects more transparently, Table A6 reports estimates from a difference-in-differences specification that replaces the full set of year fixed effects in Equation 1

<sup>12</sup>The analysis is restricted to counties that belonged to Prussia in 1849, as occupational census data for that year are only available for this subset. I end the analysis in 1939 to avoid confounding effects from the major border changes that occurred after World War II.

<sup>13</sup>Although the 1849 data point alone does not permit a formal pre-trend test, the discrete employment shift between 1849 and 1882, followed by a stable divergence through 1939, is consistent with the interpretation that these differences were caused by the onset of industrial-scale coal mining rather than pre-existing trends. I further assess robustness in Section 7.

with a single post-1849 indicator. Relative to 1849, hard coal regions experienced increases of 0.42 (log) standard deviations (SD) in total employment (approximately 37% in levels), 0.18 (log) SD in manufacturing (19.4%), and 0.14 (log) SD in services (16%). By contrast, lignite regions saw a decline of 0.19 (log) SD in manufacturing employment, equivalent to roughly 20% in levels.

These findings highlight how the strength of local production linkages shaped the long-run regional effects of extraction. In hard coal areas, strong integration into industrial supply chains supported broad-based growth. In lignite regions, by contrast, weak linkages limited spillovers and contributed to structural decline.

## 6.2 Effects of Resource Booms: Reduced-Form Shift-Share Estimates

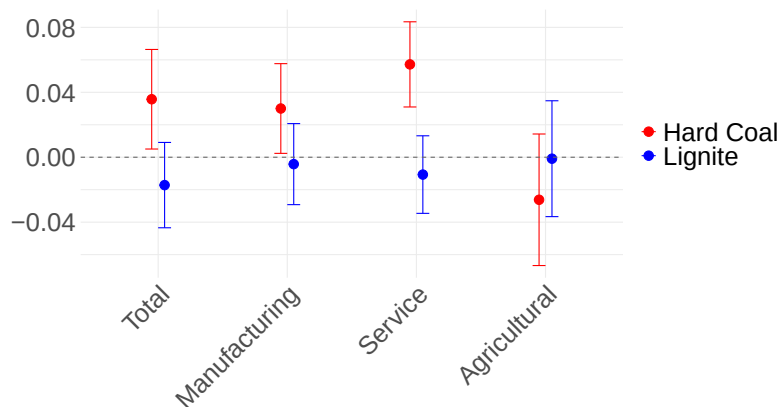


Figure 5: Reduced-form shift-share regressions: Effect of Resource Booms on Sectoral Employment (1882–1939)

*Note:* This figure plots coefficients and 95% confidence intervals from reduced-form shift-share regressions based on Equation 2. Each estimate captures the differential effect of changes in global coal prices on counties with hard coal (red) and lignite (blue) endowments, relative to non-endowed counties. The dependent variables are county-level employment in total, manufacturing, services, and agriculture. All outcomes are log-transformed and standardized to have a mean of zero and a standard deviation of one. Standard errors are clustered at the county level. The analysis covers the period 1882–1939 within the borders of the German Empire.

I now turn to the reduced-form shift-share estimates from Equation 2, which complement the event-study results by exploiting time-series variation in global coal prices, interacted with local geological endowments. This approach allows me to estimate how county-level labor markets responded to coal booms, holding fixed local characteristics and national trends.

Figure 5 presents the estimated effects of global coal price shocks on sectoral employment in counties with hard coal (red) and lignite (blue) deposits, relative to non-endowed counties. The

results closely mirror the patterns from the event-study analysis.

An increase in global coal prices leads to a statistically significant rise in total employment in hard coal regions, reflecting increased labor demand during global booms. These gains are concentrated in manufacturing and services, supporting the hypothesis that hard coal generated strong upstream and downstream linkages as well as local demand spillovers. In contrast, agricultural employment declines slightly in hard coal regions, consistent with ongoing industrialization, though the effect is not statistically significant. In lignite regions, by contrast, price shocks are associated with no significant changes in any sector. This supports the interpretation that lignite was weakly integrated into the local economy and failed to generate meaningful production or demand-side spillovers.

While the direction of the effects aligns with the event-study results, the estimated magnitudes are notably smaller. This is consistent with the different nature of the designs: the shift-share specification captures marginal, short-run effects of global price shocks, whereas the event-study estimates reflect the long-run cumulative impact of resource extraction. My results show that while global price fluctuations do affect local labor markets, the full impact of extraction unfolds over a longer horizon and through broader structural transformations.

Taken together, these findings provide further evidence that hard coal extraction supported broad-based industrial employment, while lignite extraction did not trigger similar economic gains. By leveraging an alternative identification strategy, the shift-share estimates reinforce the robustness of the main results and underscore the importance of production linkages in shaping local responses to global commodity cycles.

### **6.3 Effects of Resource Phase-Out**

So far, I have shown that hard coal mining generated significant economic benefits for producer regions during the extraction period. But what happens when extraction gradually declines? Do former mining regions retain an industrial advantage through agglomeration forces, or do these benefits erode once extraction ceases?

To examine this, I re-estimate the event-study specification from Equation 1, using 1961, the onset of the hard coal phase-out, as the baseline year. Figure 6 plots coefficients for total, manufacturing, service, and agricultural employment, comparing hard coal and lignite regions to non-coal

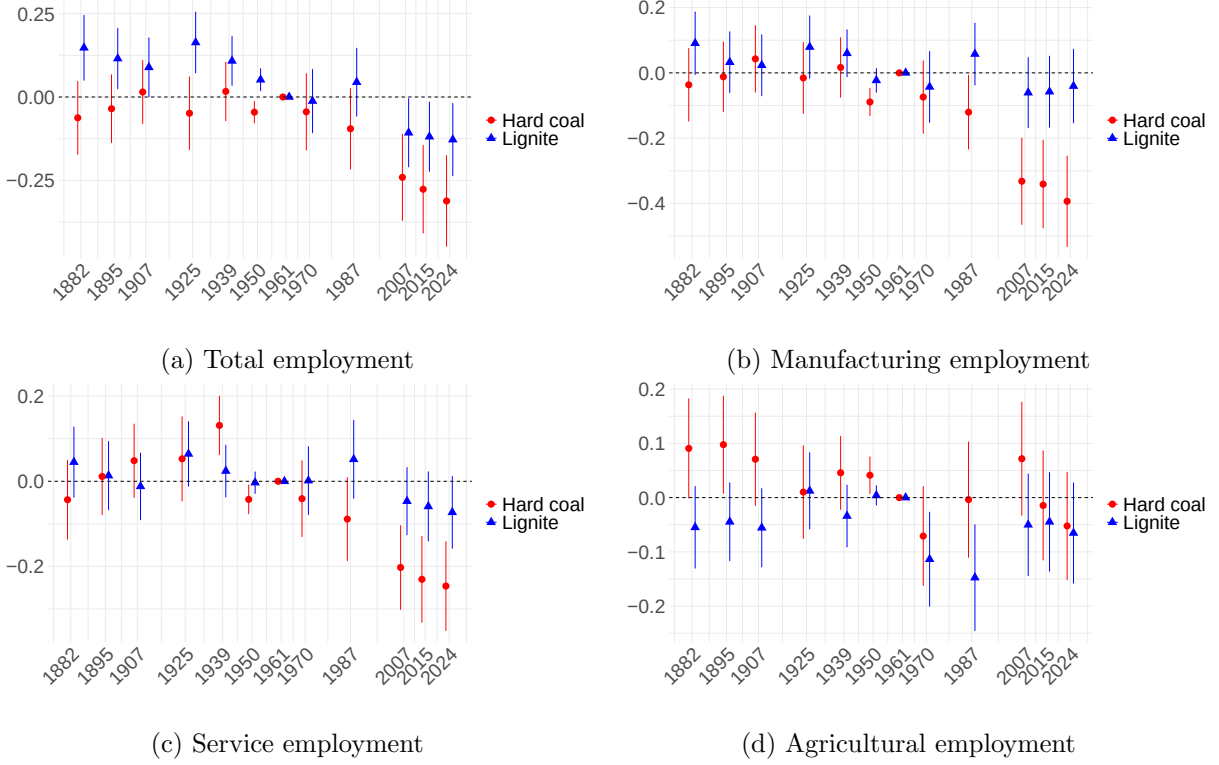


Figure 6: Event-Study: Effect of Coal Phase-Out on Sectoral Employment (1882–2024)

*Note:* This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a standard deviation of one. Hard coal and lignite indicators are interacted with year fixed effects; the omitted year is 1961. Standard errors are clustered at the county level. The analysis is restricted to West German counties.

regions in West Germany over time. The results indicate that the economic advantages associated with hard coal production did not persist. In manufacturing (Panel 6b), I find a clear and persistent decline in employment in hard coal regions relative to non-coal regions following 1961. The negative trend suggests that the industrial base in these regions eroded alongside the contraction of coal mining. By contrast, I find no significant employment changes in lignite regions. Estimates remain close to zero and are statistically insignificant. This is consistent with earlier findings that lignite mining was only weakly integrated into the local production structure, implying fewer negative effects once production declined.

The other panels of Figure 6 show similar sectoral patterns. In the service sector (Panel 6c), hard coal regions experience a relative decline in employment, likely driven by falling population

and income levels, which weaken local demand linkages. I also observe modest service employment reductions in lignite regions starting around 2007, likely reflecting the gradual reduction in mining employment and associated demand spillovers.

In the agricultural sector (Panel 6d), I find little evidence of significant employment changes in either type of coal region relative to non-coal areas. In total employment (Panel 6a), hard coal regions exhibit a persistent decline following the phase-out. Lignite regions also show a downward trend, driven by service sector reductions, though the magnitude is considerably smaller.

To assess the magnitude of the long-run effects of the coal phase-out, Table A7 reports results from a difference-in-differences regression using manufacturing employment as the outcome. The specification replaces year fixed effects with a single post-1961 indicator. The estimates indicate that manufacturing employment in hard coal regions declined by 0.3 (log) standard deviations relative to non-coal regions. In levels, this corresponds to an average decline of approximately 31%, suggesting that the phase-out had substantial and persistent effects on local industrial employment.

Taken together, these findings show that the economic benefits of hard coal production eroded as mining activities declined. Rather than leaving behind a lasting industrial advantage, former producer regions experienced a relative contraction in employment in manufacturing and service.

## 6.4 Understanding the Divergent Economic Effects of Hard Coal and Lignite Mining: The Importance of Linkages

To better understand the mechanisms behind the divergent economic effects of hard coal and lignite documented in the previous section, I disaggregate the manufacturing sector by the degree of upstream or downstream linkage to coal mining (see Section 4.2.1). For clarity and comparability, I estimate a difference-in-difference (DiD) specification based on Equation 1, replacing the full set of year indicators with a single post-treatment dummy. This enables a direct comparison of average employment changes across industries with varying degrees of production linkages to mining.

The left panel of Figure 7 shows estimates for the extraction period (1849–1939). In hard coal regions, manufacturing employment rose significantly following the onset of large-scale coal mining, with gains concentrated in industries closely linked to coal through input–output relationships. Non-linked industries saw smaller and statistically insignificant gains. By contrast, lignite regions experienced overall declines in manufacturing employment, with slightly smaller losses in linked

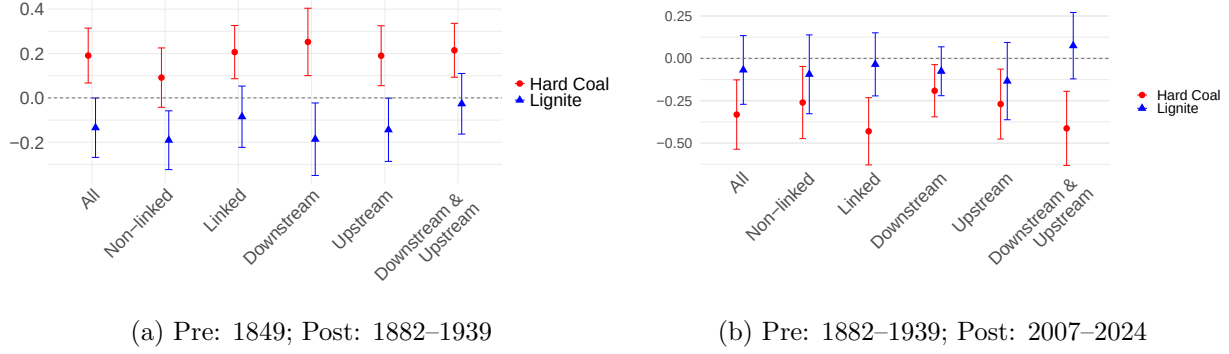


Figure 7: Difference-in-Differences regressions: Effect of Coal Endowment on Disaggregated Manufacturing Employment

*Note:* This figure plots coefficients and 95% confidence intervals from difference-in-differences regressions based on Equation 1, where the full set of year fixed effects is replaced by a single post-period indicator. Both panels report estimated effects of hard coal (red) and lignite (blue) endowments on county-level manufacturing employment, disaggregated by linkage to coal mining. All outcomes are log-transformed and standardized to have a mean of zero and a standard deviation of one. The left panel covers the extraction period (1849–1939) using counties within Prussian borders. The post-period indicator equals one for all years after 1849. The right panel shows results for the coal phase-out period (1882–2024) in West Germany, where the post-period indicator equals one for the years 2007, 2015, 2024. Standard errors are clustered at the county level.

industries. This pattern suggests that lignite mining did not generate substantial manufacturing linkages and was instead associated with local industrial decline. Disaggregating further, the positive employment effects in hard coal regions are observed across all types of linked industries, those connected to mining through upstream linkages, downstream linkages, or both. This suggests that hard coal mining supported both local input suppliers and coal-using industries. In lignite regions, by contrast, employment effects remain weak or negative across all linkage types. For industries linked both upstream and downstream, cost pressures and potential spillovers appear to offset each other, resulting in no net gain. For those with only one type of linkage, negative effects dominate, consistent with crowding-out from rising wages or land costs and limited spillovers.

The right panel of Figure 7 presents estimates for the coal phase-out period (1882–2024). Here, the earlier patterns reverse. In hard coal regions, the sharpest employment declines occur in industries that had previously been linked to mining, especially those with both upstream and downstream linkages. These industries, tightly integrated into the coal-based economy, appear particularly vulnerable to the sector’s contraction. Non-linked industries also experienced significant, though smaller, declines. In lignite regions, by contrast, there are no significant differences across industry groups. This is consistent with lignite’s weaker integration into the surrounding industrial

base.

**Sectoral Concentration of Manufacturing Employment.** A potential alternative explanation for the sharp decline in hard coal regions following the phase-out is that these areas had developed narrow, undiversified manufacturing structures, leaving them exposed to sector-specific shocks. To assess this, I examine changes in local industry concentration using the Herfindahl-Hirschman Index of manufacturing employment (see Section A.3). As shown in Figure A9, the opposite pattern emerges: during the extraction period, hard coal regions became more diversified than non-coal areas. This likely reflects the broad input-output linkages of hard coal mining, which supported a wide range of supplier and user industries. By contrast, lignite regions saw slightly increasing industrial concentration over time. These results reinforce the interpretation that the vulnerability of hard coal regions was not driven by a narrow industrial base, but by concentrated dependence on a single lead sector. It was the depth of integration with coal, rather sectoral concentration of manufacturing employment, that amplified long-run exposure to structural decline.

Taken together, the findings reinforce that hard coal mining stimulated local industrial growth through strong upstream and downstream linkages. Lignite mining, by contrast, failed to generate meaningful spillovers in either direction. However, the same production linkages that supported growth in hard coal regions also amplified their exposure to decline when mining contracted, particularly in industries doubly connected to the coal sector. Importantly, the vulnerability observed in these regions was not due to low industrial diversification: hard coal regions became more diversified during the extraction period. Instead, it was the depth of integration with the coal economy, rather than sectoral concentration per se, that shaped both the benefits during extraction and the risks during decline.

## 7 Robustness of the Results

To demonstrate that my results are not sensitive to the specific choice of variables, sample, or models, I perform several robustness checks.

## 7.1 Alternative Definitions of Coal Exposure

A first concern is that the estimated effects may be sensitive to how exposure to coal deposits is defined. In the baseline specification, I classify a county as exposed to hard coal or lignite if a deposit lies within a 20 kilometer radius of the county centroid. To test the robustness of the geographic treatment definition, I re-estimate the difference-in-differences specification based on Equation 1 for the coal extraction period (1849-1939), using manufacturing employment as the outcome and replacing year fixed effects with a single post-period indicator. I vary the distance threshold for coal endowment from the baseline 20 kilometers to alternative cutoffs of 10, 15, 25, and 30 kilometers. As shown in Table A9, the results remain qualitatively consistent across specifications. The baseline 20-kilometer radius neither maximizes the estimated effect of hard coal nor minimizes the negative effect of lignite. Additionally, I restrict the analysis to coal deposits located within Prussia. The results, reported in Column (2) of Table A8, remain robust. Finally, I replace the binary treatment indicators with continuous measures of proximity to the nearest hard coal or lignite deposit, allowing treatment intensity to vary with distance. As shown in Column (3) of Table A8, the results align with the baseline: manufacturing employment increases with proximity to hard coal and decreases with proximity to lignite. Overall, these findings show that the main results are not driven by the specific geographic definition of coal exposure and hold across a range of alternative treatment measures.

## 7.2 Potential Confounding Factors

Table A10 reports results from a series of difference-in-differences regressions based on Equation 1 for the coal extraction period (1849–1939), using manufacturing employment as the outcome. The specification replaces year fixed effects with a single post-period indicator, and additional controls are sequentially added to assess robustness. Column (2) includes log distance to London to capture regional variation in exposure to technological diffusion from Britain. Column (3) adds distance to Wittenberg, a commonly used proxy for historical Protestant influence, linked to differences in human capital and institutional development (Becker & Woessmann, 2009). Column (4) includes distance to the Polish border to account for possible variation in labor supply and cross-border migration. Column (5) controls for log distance to Berlin, reflecting potential advantages from

proximity to the capital such as access to political influence or public investment. Column (6) adds a west-of-Elbe indicator to capture Germany’s historical east–west divide in agrarian structure and labor institutions (Tilly & Kopsidis, 2020). Finally, Column (7) includes a dummy for counties located in territories formerly occupied by France, accounting for long-run institutional effects of Napoleonic reforms (Acemoglu et al., 2011). Across all specifications, the estimated effects of coal exposure on manufacturing employment remain remarkably stable in both magnitude and statistical significance. This reinforces the interpretation that the main results are not confounded by pre-existing regional differences in geography, institutions, or human capital.

### 7.3 Potential Contamination Bias

To address concerns that including both coal types in a single regression may blur their distinct effects, I re-estimate the event-study model from Equation 1 separately for hard coal and lignite endowments. Figures A7 and A8 plot the coefficients from these separate specifications. The results closely mirror the baseline estimates reported in Figure 4, confirming that the observed effects are not driven by contamination across treatment types. As a further robustness check, I exclude all counties containing both hard coal and lignite deposits. Table A5 presents event-study estimates based on this restricted sample. The results remain consistent in both magnitude and significance, suggesting that the main findings are not driven by overlapping exposure.

### 7.4 Sensitivity to Outlier Regions

To ensure that the results are not driven by a disproportionately large coal-producing district, I re-estimate the main specifications excluding key outliers. Specifically, I drop the largest hard coal-producing district (Arnsberg) and the largest lignite-producing district (Merseburg), based on total production between 1850 and 1914 (own calculations using data from Kunz (2008)). As shown in Columns (2) and (4) of Table A11, the results remain robust in both magnitude and significance, indicating that the main findings are not driven by outlier regions.

### 7.5 Sensitivity to Timing of Extraction

A potential concern is that the stronger economic effects observed in hard coal regions may reflect their earlier onset of extraction rather than structural differences in linkage intensity (see

Figure A1a). To address this, I re-estimate the main event-study specifications excluding Oppeln, the earliest major hard coal-producing district based on production levels in 1850 (own calculations using data from Kunz (2008)). As shown in Column (3) of Table A11, the results remain robust in both magnitude and statistical significance. For lignite, Merseburg was both the earliest and the largest producer over the period; corresponding results excluding this district are reported in Column (4). These findings, together with the evidence that employment gains in hard coal regions were concentrated in sectors with strong upstream and downstream linkages, support the interpretation that the observed divergence reflects differences in economic integration, rather than differences in timing of extraction.

## 7.6 Balanced Subsamples

A potential concern is that changes in county boundaries over time could bias results if sample composition is systematically related to coal endowments. To address this, I re-estimate the event-study specifications (Equation 1) using a balanced panel of counties consistently observed across the full time period. Figure A4 presents results for the extraction period (1849–1939). While standard errors are somewhat larger, the direction and magnitude of the effects remain consistent with the baseline estimates from Figure 4: hard coal regions exhibit persistent increases in total, manufacturing, and service employment, while lignite regions show long-run declines. Figure A5 shows results for the coal phase-out period (1882–2024) using a balanced sample. Again, the patterns closely mirror the baseline estimates from Figure 6, with hard coal regions experiencing gradual but persistent employment losses, and lignite regions showing no comparable trend. Overall, these findings confirm that the main results are not driven by sample composition or administrative boundary changes.

## 7.7 Assessing Potential Pre-trends

The event-study model in Equation 1 relies on the assumption that, absent coal endowments, regions with and without deposits would have followed similar economic trends, conditional on fixed effects and observed controls. The discrete jump in employment after 1849, followed by stable post-treatment differences, already provides support to this assumption. To further assess its validity, I test for differential pre-trends using an alternative outcome: wages. Specifically, I

re-estimate Equation 1 using unskilled wages as the outcome variable and control for local price levels to account for cost-of-living differences (see Section A.5 in the Appendix for data sources). Figure A6 plots the estimated effects of hard coal and lignite endowments on real wages from 1800 to 1914. The results show no large differences in wage levels or trends between coal and non-coal regions prior to 1849, supporting the parallel trends assumption. Starting in the mid-19th century, wages begin to rise in hard coal regions, consistent with increasing labor demand driven by expanding mining and industrial activity. These patterns are consistent with demand-side effects from resource extraction and reinforce the interpretation that the observed employment trends are not driven by differential pre-treatment dynamics.

## 7.8 Alternative Definitions of the Shift Variable

To assess the robustness of the shift-share estimates, I test whether the results are sensitive to alternative definitions of the time-varying "shift" component. In the baseline specification, the shift is given by the world coal price, which captures global demand and supply shocks in the coal market and is plausibly exogenous to local economic conditions in Germany. This measure avoids endogeneity concerns associated with national variables that may partly reflect local outcomes in coal-producing regions. As a robustness check, I re-estimate the shift-share model using several alternative, national-level indicators of coal sector activity: German hard coal and lignite production, prices, and revenue (price  $\times$  quantity). These alternatives allow for a differentiation between hard coal and lignite dynamics. As shown in Figure A12, the results remain consistent across specifications in both magnitude and significance. This confirms that the estimated effects are not sensitive to the precise choice of the shift variable.

To assess whether the design is spuriously capturing unrelated global shocks, I run a placebo shift-share test that replaces coal price shocks with shocks to unrelated commodities (coffee, sugar, tobacco, and rice). The estimated interaction coefficients for hard-coal regions are near zero or slightly negative (Figure A13b), providing no evidence of analogous effects and reinforcing the validity of the identification strategy.

## 7.9 Alternative Input-Output Table

To ensure that the observed heterogeneity by linkage intensity is not driven by the specific input–output table used, I replicate the analysis using an alternative historical German table from Fremdling and Staeglin (2014), based on data from 1936. Industries are reclassified by their upstream and downstream linkage intensity using this table (see Table A3), and I re-estimate the difference-in-differences models for both the coal extraction and the coal phase-out period. As shown in Figure A10 in the Appendix, the results remain highly consistent in both magnitude and significance, confirming that the findings are robust to the choice of input–output classification.

Together, these robustness checks confirm the validity of my results across both the extraction and the phase-out period.

## 8 Conclusion

In this paper, I show that the long-run local effects of natural resource extraction depend critically on the extent to which extractive industries are integrated into the regional economy. By comparing two types of coal, hard coal and lignite, extracted under similar institutional conditions in Germany, I demonstrate that the capacity of a resource sector to generate upstream and downstream production linkages is a key mechanism shaping its broader economic impact.

Regions specialized in hard coal mining, which produced strong local input–output linkages, experienced broad-based industrial growth during the extraction phase, with spillovers into both manufacturing and services. Positive spillovers to manufacturing were largely driven by industries with upstream or downstream linkages to coal mining. Importantly, these regions also became more diversified over time, suggesting that resource-driven growth does not necessarily lead to industrial concentration. However, the very linkages that supported growth during the boom years also amplified exposure to decline: once coal production contracted, these tightly connected regions faced persistent employment losses, especially in industries closely tied to mining.

By contrast, lignite regions, where extractive activity remained largely isolated from the local production structure, saw adverse effects even during the extraction period, including deindustrialization and declines in non-tradable employment. Since lignite was weakly integrated into other sectors, these regions were less exposed to economic decline during the phase-out. Still, they saw

some contraction in services, likely reflecting reduced mining employment and associated demand spillovers.

Taken together, the findings help explain why resource-rich regions may experience markedly different long-run outcomes. Resource extraction can yield lasting local benefits when it fosters strong upstream and downstream linkages with the broader economy. In regions where extractive activity remains economically isolated, stagnation or decline is more likely. This mechanism helps reconcile conflicting findings in the resource curse literature and clarifies why the local legacy of resource wealth ranges from industrial takeoff to persistent stagnation.

For policymakers, the results underscore the importance of recognizing the production linkages generated by the extraction sector in the local economy. Promoting stronger upstream and downstream integration through the development of local supply chains can enhance the potential for extractive activity to contribute to broad-based economic growth. However, such integration also amplifies vulnerabilities during downturns, making it essential to anticipate and prepare for the long-term adjustment costs that follow the end of extraction.

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## A Appendix

### A.1 Additional Tables

#### A.1.1 Descriptives

Table A1: Descriptive Statistic: Cross-Section

| Variable                             | Mean  | Median | Sd   | Min | Max |
|--------------------------------------|-------|--------|------|-----|-----|
| Distance Hard coal (in km)           | 132   | 105    | 111  | 0   | 552 |
| Distance Lignite (in km)             | 110   | 71     | 110  | 0   | 566 |
| Hard coal region (dummy)             | 0.16  | 0      | 0.36 | 0   | 1   |
| Lignite region (dummy)               | 0.18  | 0      | 0.39 | 0   | 1   |
| Hard coal or lignite region (dummy)  | 0.3   | 0      | 0.46 | 0   | 1   |
| Hard coal and lignite region (dummy) | 0.034 | 0      | 0.18 | 0   | 1   |

Table A2: Upstream and Downstream Linkage Measures for U.S. Input–Output Sectors (1947)

| Industry                                              | Upstream |          | Downstream |          | Linked     |          |       |
|-------------------------------------------------------|----------|----------|------------|----------|------------|----------|-------|
|                                                       | Direct   | Indirect | Direct     | Indirect | Downstream | Upstream | Total |
| <b>Apparel and leather and allied products</b>        | 0.0      | 0.2      | 0.1        | 0.6      | No         | No       | No    |
| <b>Chemical products</b>                              | 1.5      | 0.4      | 1.8        | 1.2      | No         | Yes      | Yes   |
| <b>Fabricated metal products</b>                      | 1.0      | 0.7      | 1.7        | 5.8      | Yes        | Yes      | Yes   |
| <b>Food and beverage and tobacco products</b>         | 0.0      | 0.2      | 0.1        | 0.6      | No         | No       | No    |
| <b>Machinery</b>                                      | 2.9      | 0.7      | 0.3        | 3.7      | Yes        | Yes      | Yes   |
| <b>Miscellaneous manufacturing</b>                    | 0.1      | 0.6      | 2.9        | 2.6      | Yes        | No       | Yes   |
| <b>Motor vehicles, bodies and trailers, and parts</b> | 0.7      | 0.4      | 0.2        | 2.9      | Yes        | No       | No    |
| <b>Nonmetallic mineral products</b>                   | 0.9      | 0.5      | 15.2       | 1.8      | Yes        | Yes      | Yes   |
| <b>Paper products</b>                                 | 0.3      | 0.4      | 1.9        | 1.0      | No         | No       | Yes   |
| <b>Petroleum and coal products</b>                    | 1.2      | 0.8      | 5.1        | 2.4      | Yes        | Yes      | Yes   |
| <b>Plastics and rubber products</b>                   | 0.2      | 0.9      | 0.7        | 1.2      | No         | No       | No    |
| <b>Primary metals</b>                                 | 0.8      | 0.7      | 14.4       | 1.1      | Yes        | Yes      | Yes   |
| <b>Printing and related support activities</b>        | 0.3      | 0.6      | 0.1        | 1.5      | No         | No       | No    |
| <b>Textile mills and textile product mills</b>        | 0.0      | 0.4      | 0.3        | 0.5      | No         | No       | No    |
| <b>Wood products</b>                                  | 2.1      | 0.4      | 0.4        | 1.6      | No         | Yes      | Yes   |

*Note:* This table reports direct and indirect upstream (sales to mining) and downstream (inputs from mining) linkage shares in percent for U.S. manufacturing and service sectors, based on the 1947 summary-level Use table (before redefinitions) (U.S. Bureau of Economic Analysis, 2019). I compute linkage shares on the full I-O matrix, but report only those industries that can be matched to historical occupations, and I exclude own-use (diagonal). Consequently, the reported column sums for upstream and downstream shares (direct and indirect) need not equal one. Binary “linked” indicators are defined as specified in Section 4.2.1.

Table A3: Upstream and Downstream Linkage Measures for German Input–Output Sectors (1936)

| Industry                                       | Upstream |          | Downstream |          | Linked     |          |       |
|------------------------------------------------|----------|----------|------------|----------|------------|----------|-------|
|                                                | Direct   | Indirect | Direct     | Indirect | Downstream | Upstream | Total |
| <b>Building and construction</b>               | 0.3      | 1.3      | 0.4        | 2.8      | No         | Yes      | Yes   |
| <b>Chemical industry</b>                       | 2.4      | 0.4      | 4.6        | 5.2      | Yes        | Yes      | Yes   |
| <b>Chemical-technical industry</b>             | 0.0      | 1.2      | 0.0        | 4.1      | No         | No       | Yes   |
| <b>Clothing</b>                                | 0.0      | 1.3      | 0.0        | 1.1      | No         | Yes      | Yes   |
| <b>Edible oil and fats</b>                     | 0.0      | 0.1      | 0.0        | 2.0      | No         | No       | No    |
| <b>Electrical engineering</b>                  | 0.4      | 1.0      | 0.5        | 4.0      | Yes        | Yes      | Yes   |
| <b>Electricity, gas and water</b>              | 4.0      | 0.7      | 33.4       | 3.0      | Yes        | Yes      | Yes   |
| <b>Fabricated iron and steel products</b>      | 0.6      | 0.9      | 0.0        | 6.8      | Yes        | Yes      | Yes   |
| <b>Food, beverages and tobacco</b>             | 0.0      | 0.7      | 0.1        | 2.4      | No         | No       | Yes   |
| <b>Foundries</b>                               | 6.4      | 0.4      | 0.0        | 6.8      | Yes        | Yes      | Yes   |
| <b>Fuel industries</b>                         | 0.0      | 1.2      | 18.8       | 2.6      | Yes        | No       | Yes   |
| <b>Glass</b>                                   | 0.0      | 0.6      | 0.0        | 10.2     | Yes        | No       | No    |
| <b>Leather industry</b>                        | 0.0      | 0.2      | 0.0        | 2.0      | No         | No       | No    |
| <b>Machinery</b>                               | 0.0      | 1.6      | 0.0        | 4.4      | No         | Yes      | Yes   |
| <b>Manufacture of paper and paper products</b> | 0.0      | 0.4      | 0.0        | 5.2      | Yes        | No       | No    |
| <b>Manufactured wood products</b>              | 0.2      | 0.6      | 0.1        | 2.5      | No         | No       | Yes   |
| <b>Metal products</b>                          | 0.0      | 1.5      | 0.0        | 5.5      | Yes        | Yes      | Yes   |
| <b>Non-ferrous metals</b>                      | 0.9      | 0.5      | 4.8        | 4.1      | Yes        | Yes      | Yes   |
| <b>Precision engineering, optics</b>           | 0.0      | 1.2      | 0.0        | 3.5      | No         | No       | No    |
| <b>Printing and duplicating</b>                | 0.4      | 0.8      | 0.0        | 1.8      | No         | No       | No    |
| <b>Rubber and asbestos manufacturing</b>       | 0.0      | 1.4      | 0.0        | 5.9      | Yes        | Yes      | Yes   |
| <b>Saw mills, timber processing</b>            | 12.9     | 0.2      | 0.0        | 2.8      | No         | Yes      | Yes   |
| <b>Spirits industry</b>                        | 0.0      | 0.5      | 0.0        | 1.8      | No         | No       | No    |
| <b>Stone and quarrying</b>                     | 0.1      | 0.8      | 3.5        | 13.7     | Yes        | No       | Yes   |
| <b>Textiles</b>                                | 0.1      | 0.1      | 0.0        | 2.8      | No         | No       | No    |

*Note:* This table reports direct and indirect upstream (sales to mining) and downstream (inputs from mining) linkage shares in percent for German manufacturing sectors, based on the 1936 I-O table (Fremdling & Staeglin, 2014). I compute linkage shares on the full I-O matrix, but report only those industries that can be matched to historical occupations, and I exclude own-use (diagonal). Consequently, the reported column sums for upstream and downstream shares (direct and indirect) need not equal one. Binary “linked” indicators are defined as specified in Section 4.2.1.

### A.1.2 Event-studies

Table A4: Event-Study: Effect of Resource Endowment on Sectoral Employment (1849–1939)

|                         | Total<br>(1)         | Manufacturing<br>(2) | Service<br>(3)       | Agricultural<br>(4) |
|-------------------------|----------------------|----------------------|----------------------|---------------------|
| HardCoal x 1882         | 0.390***<br>(0.080)  | 0.154***<br>(0.056)  | 0.060<br>(0.064)     | 0.036<br>(0.076)    |
| HardCoal x 1895         | 0.417***<br>(0.086)  | 0.178***<br>(0.062)  | 0.119*<br>(0.069)    | 0.047<br>(0.072)    |
| HardCoal x 1907         | 0.468***<br>(0.093)  | 0.232***<br>(0.067)  | 0.159**<br>(0.073)   | 0.005<br>(0.083)    |
| HardCoal x 1925         | 0.395***<br>(0.098)  | 0.168**<br>(0.076)   | 0.155**<br>(0.079)   | -0.082<br>(0.086)   |
| HardCoal x 1939         | 0.463***<br>(0.100)  | 0.202***<br>(0.078)  | 0.241***<br>(0.082)  | -0.031<br>(0.080)   |
| Lignite x 1882          | -0.175**<br>(0.076)  | -0.143**<br>(0.062)  | -0.161***<br>(0.062) | 0.243***<br>(0.083) |
| Lignite x 1895          | -0.207**<br>(0.081)  | -0.200***<br>(0.067) | -0.195***<br>(0.066) | 0.258***<br>(0.081) |
| Lignite x 1907          | -0.234***<br>(0.085) | -0.209***<br>(0.072) | -0.222***<br>(0.070) | 0.242***<br>(0.091) |
| Lignite x 1925          | -0.157*<br>(0.088)   | -0.152**<br>(0.075)  | -0.142**<br>(0.070)  | 0.348***<br>(0.092) |
| Lignite x 1939          | -0.215**<br>(0.091)  | -0.170**<br>(0.077)  | -0.184**<br>(0.074)  | 0.271***<br>(0.091) |
| Geo Controls            | Yes                  | Yes                  | Yes                  | Yes                 |
| County fixed effects    | Yes                  | Yes                  | Yes                  | Yes                 |
| Year fixed effects      | Yes                  | Yes                  | Yes                  | Yes                 |
| Observations            | 5,834                | 5,834                | 5,834                | 5,834               |
| Adjusted R <sup>2</sup> | 0.89                 | 0.91                 | 0.90                 | 0.86                |

*Notes:* This table reports coefficients from event-study regressions based on Equation 1. The dependent variables are the log of standardized county-level employment in (1) total, (2) manufacturing, (3) service, and (4) agricultural sectors (mean zero, unit variance). Each row presents estimated effects for counties with hard coal or lignite deposits interacted with year fixed effects. the omitted year is 1849. All regressions include geographic controls, county fixed effects, and year fixed effects. Standard errors, clustered at the county level, are shown in parentheses. The analysis is restricted to Prussian counties. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

### A.1.3 Difference-in-Difference Regressions

Table A5: Event-Study: Excluding Overlapping Coal Regions (1849–1939)

|                         | Total<br>(1)        | Manufacturing<br>(2) | Service<br>(3)      | Agricultural<br>(4) |
|-------------------------|---------------------|----------------------|---------------------|---------------------|
| HardCoal x 1882         | 0.468***<br>(0.093) | 0.163**<br>(0.064)   | 0.088<br>(0.075)    | 0.028<br>(0.089)    |
| HardCoal x 1895         | 0.506***<br>(0.100) | 0.207***<br>(0.071)  | 0.161**<br>(0.081)  | 0.024<br>(0.083)    |
| HardCoal x 1907         | 0.559***<br>(0.108) | 0.255***<br>(0.077)  | 0.212**<br>(0.087)  | -0.019<br>(0.093)   |
| HardCoal x 1925         | 0.474***<br>(0.114) | 0.162*<br>(0.089)    | 0.212**<br>(0.094)  | -0.101<br>(0.097)   |
| HardCoal x 1939         | 0.518***<br>(0.118) | 0.175*<br>(0.092)    | 0.268***<br>(0.099) | -0.018<br>(0.088)   |
| Lignite x 1882          | -0.064<br>(0.077)   | -0.119*<br>(0.066)   | -0.109*<br>(0.066)  | 0.225**<br>(0.092)  |
| Lignite x 1895          | -0.087<br>(0.082)   | -0.160**<br>(0.071)  | -0.131*<br>(0.069)  | 0.229**<br>(0.092)  |
| Lignite x 1907          | -0.112<br>(0.085)   | -0.175**<br>(0.076)  | -0.150**<br>(0.073) | 0.210**<br>(0.101)  |
| Lignite x 1925          | -0.045<br>(0.087)   | -0.136*<br>(0.079)   | -0.068<br>(0.072)   | 0.315***<br>(0.102) |
| Lignite x 1939          | -0.119<br>(0.091)   | -0.169**<br>(0.081)  | -0.130*<br>(0.076)  | 0.261**<br>(0.101)  |
| Geo Controls            | Yes                 | Yes                  | Yes                 | Yes                 |
| County fixed effects    | Yes                 | Yes                  | Yes                 | Yes                 |
| Year fixed effects      | Yes                 | Yes                  | Yes                 | Yes                 |
| Observations            | 5,622               | 5,622                | 5,622               | 5,622               |
| Adjusted R <sup>2</sup> | 0.89                | 0.91                 | 0.90                | 0.86                |

*Notes:* This table reports coefficients from event-study regressions based on Equation 1. Counties with overlapping deposits of hard coal and lignite are excluded from the analysis. The dependent variables are the log of standardized county-level employment in (1) total, (2) manufacturing, (3) service, and (4) agricultural sectors (mean zero, unit variance). Each row presents estimated effects for counties with hard coal or lignite deposits interacted with year fixed effects. the omitted year is 1849. All regressions include geographic controls, county fixed effects, and year fixed effects. Standard errors, clustered at the county level, are shown in parentheses. The analysis is restricted to Prussian counties. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

Table A6: Difference-in-Difference Regressions: Effect of Resource Endowment on Sectoral Employment (1849–1939)

|                         | Total<br>(1)         | Manufacturing<br>(2) | Service<br>(3)       | Agricultural<br>(4) |
|-------------------------|----------------------|----------------------|----------------------|---------------------|
| HardCoal x Post1849     | 0.420***<br>(0.087)  | 0.180***<br>(0.064)  | 0.139**<br>(0.069)   | -0.008<br>(0.073)   |
| Lignite x Post1849      | -0.211***<br>(0.081) | -0.187***<br>(0.068) | -0.195***<br>(0.065) | 0.262***<br>(0.083) |
| Geo Controls            | Yes                  | Yes                  | Yes                  | Yes                 |
| County fixed effects    | Yes                  | Yes                  | Yes                  | Yes                 |
| Year fixed effects      | Yes                  | Yes                  | Yes                  | Yes                 |
| Observations            | 5,834                | 5,834                | 5,834                | 5,834               |
| Adjusted R <sup>2</sup> | 0.89                 | 0.90                 | 0.90                 | 0.85                |

*Notes:* This table reports coefficients from difference-in-differences regressions based on Equation 1, where the full set of year fixed effects is replaced by a single post-period indicator. The dependent variables are the log of standardized county-level employment in (1) total, (2) manufacturing, (3) services, and (4) agriculture (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls, county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

Table A7: Difference-in-Difference Regressions: Effect of Coal Phase-Out on Sectoral Employment (1882–2024)

|                         | Total<br>(1)         | Manufacturing<br>(2) | Service<br>(3)       | Agricultural<br>(4) |
|-------------------------|----------------------|----------------------|----------------------|---------------------|
| HardCoal x Post1961     | -0.216***<br>(0.060) | -0.298***<br>(0.061) | -0.224***<br>(0.051) | -0.047<br>(0.040)   |
| Lignite x Post1961      | -0.201***<br>(0.050) | -0.103*<br>(0.054)   | -0.070<br>(0.043)    | -0.036<br>(0.039)   |
| Geo Controls            | Yes                  | Yes                  | Yes                  | Yes                 |
| County fixed effects    | Yes                  | Yes                  | Yes                  | Yes                 |
| Year fixed effects      | Yes                  | Yes                  | Yes                  | Yes                 |
| Observations            | 10,605               | 10,605               | 10,605               | 10,588              |
| Adjusted R <sup>2</sup> | 0.85                 | 0.85                 | 0.89                 | 0.91                |

*Notes:* This table reports coefficients from difference-in-differences regressions based on Equation 1, where the full set of year fixed effects is replaced by a single post-period indicator. The dependent variables are the log of standardized county-level employment in (1) total, (2) manufacturing, (3) services, and (4) agriculture (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits interacted with a post indicator equal to one for all years after 1961. All regressions include geographic controls, county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to West German counties. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

Table A8: Difference-in-Difference Regressions: Alternative Definitions of the Treatment Variable (1849–1939)

|                         | Baseline             | Coal<br>Inside<br>Prussia | Proximity            |
|-------------------------|----------------------|---------------------------|----------------------|
|                         | (1)                  | (2)                       | (3)                  |
| HardCoal x Post1849     | 0.180***<br>(0.064)  | 0.161**<br>(0.064)        | 0.050*<br>(0.028)    |
| Lignite x Post1849      | -0.187***<br>(0.068) | -0.178***<br>(0.068)      | -0.128***<br>(0.024) |
| Geo Controls            | Yes                  | Yes                       | Yes                  |
| County fixed effects    | Yes                  | Yes                       | Yes                  |
| Year fixed effects      | Yes                  | Yes                       | Yes                  |
| Observations            | 5,834                | 5,834                     | 5,834                |
| Adjusted R <sup>2</sup> | 0.90                 | 0.90                      | 0.90                 |

*Notes:* This table reports coefficients from difference-in-differences regressions based on Equation 1, where the full set of year fixed effects is replaced by a single post-period indicator. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties classified as exposed to hard coal or lignite, with exposure defined in different ways, interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls, as well as county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$ .

Table A9: Difference-in-Difference Regressions: Varying the Distance Cut-off (1849–1939)

|                         | 10km<br>(1)          | 15km<br>(2)          | 20km<br>(3)          | 25km<br>(4)          | 30km<br>(5)          |
|-------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| HardCoal x Post1849     | 0.210***<br>(0.058)  | 0.118**<br>(0.060)   | 0.180***<br>(0.064)  | 0.102<br>(0.063)     | 0.037<br>(0.064)     |
| Lignite x Post1849      | -0.199***<br>(0.068) | -0.212***<br>(0.067) | -0.187***<br>(0.068) | -0.245***<br>(0.070) | -0.252***<br>(0.065) |
| Geo Controls            | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| County fixed effects    | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Year fixed effects      | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Observations            | 5,834                | 5,834                | 5,834                | 5,834                | 5,834                |
| Adjusted R <sup>2</sup> | 0.90                 | 0.90                 | 0.90                 | 0.90                 | 0.90                 |

*Notes:* This table reports coefficients from difference-in-differences regressions based on Equation 1, where the full set of year fixed effects is replaced by a single post-period indicator. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits, defined using varying distance cutoffs, interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls, county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

Table A10: Difference-in-Difference Regressions: Potential Confounders (1849–1939)

|                         | Baseline<br>(1)      | Distance London<br>(2) | Distance Wittenberg<br>(3) | Distance Poland<br>(4) | Distance Berlin<br>(5) | West Elbe<br>(6)    | French Occupation<br>(7) |
|-------------------------|----------------------|------------------------|----------------------------|------------------------|------------------------|---------------------|--------------------------|
| HardCoal x Post1849     | 0.180***<br>(0.064)  | 0.098<br>(0.060)       | 0.102*<br>(0.059)          | 0.105*<br>(0.058)      | 0.140**<br>(0.059)     | 0.163***<br>(0.058) | 0.165***<br>(0.058)      |
| Lignite x Post1849      | -0.187***<br>(0.068) | -0.074<br>(0.062)      | -0.132*<br>(0.079)         | -0.138*<br>(0.079)     | -0.087<br>(0.068)      | -0.141**<br>(0.069) | -0.139**<br>(0.070)      |
| Geo Controls            | Yes                  | Yes                    | Yes                        | Yes                    | Yes                    | Yes                 | Yes                      |
| County fixed effects    | Yes                  | Yes                    | Yes                        | Yes                    | Yes                    | Yes                 | Yes                      |
| Year fixed effects      | Yes                  | Yes                    | Yes                        | Yes                    | Yes                    | Yes                 | Yes                      |
| Observations            | 5,834                | 5,834                  | 5,834                      | 5,834                  | 5,834                  | 5,834               | 5,834                    |
| Adjusted R <sup>2</sup> | 0.90                 | 0.90                   | 0.90                       | 0.90                   | 0.91                   | 0.91                | 0.91                     |

*Notes:* This table reports coefficients from difference-in-differences regressions based on Equation 1, where the full set of year fixed effects is replaced by a single post-period indicator. The dependent variable is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls, county and year fixed effects. Potential confounders are added sequentially. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

Table A11: Difference-in-Difference Regressions: Sensitivity to Outlier Regions (1849–1939)

|                         | Baseline             | Largest Producer<br>Hard coal | First Mover<br>Hard Coal | First Mover =<br>Largest Producer Lignite |
|-------------------------|----------------------|-------------------------------|--------------------------|-------------------------------------------|
|                         | (1)                  | (2)                           | (3)                      | (4)                                       |
| HardCoal x Post1849     | 0.180***<br>(0.064)  | 0.113*<br>(0.066)             | 0.131**<br>(0.066)       | 0.142**<br>(0.069)                        |
| Lignite x Post1849      | -0.187***<br>(0.068) | -0.170**<br>(0.070)           | -0.171**<br>(0.068)      | -0.221***<br>(0.074)                      |
| Geo Controls            | Yes                  | Yes                           | Yes                      | Yes                                       |
| County fixed effects    | Yes                  | Yes                           | Yes                      | Yes                                       |
| Year fixed effects      | Yes                  | Yes                           | Yes                      | Yes                                       |
| Observations            | 5,834                | 5,696                         | 5,699                    | 5,726                                     |
| Adjusted R <sup>2</sup> | 0.90                 | 0.90                          | 0.91                     | 0.90                                      |

*Notes:* This table reports coefficients from difference-in-differences regressions based on Equation 1, where the full set of year fixed effects is replaced by a single post-period indicator. The dependent variables is the log of standardized county-level employment in manufacturing (mean zero, unit variance). Each row shows estimated effects for counties with hard coal or lignite deposits interacted with a post indicator equal to one for all years after 1849. All regressions include geographic controls, county and year fixed effects. Standard errors, clustered at the county level, are in parentheses. The sample is restricted to Prussian counties. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

## A.2 Additional Figures

### A.2.1 Descriptives

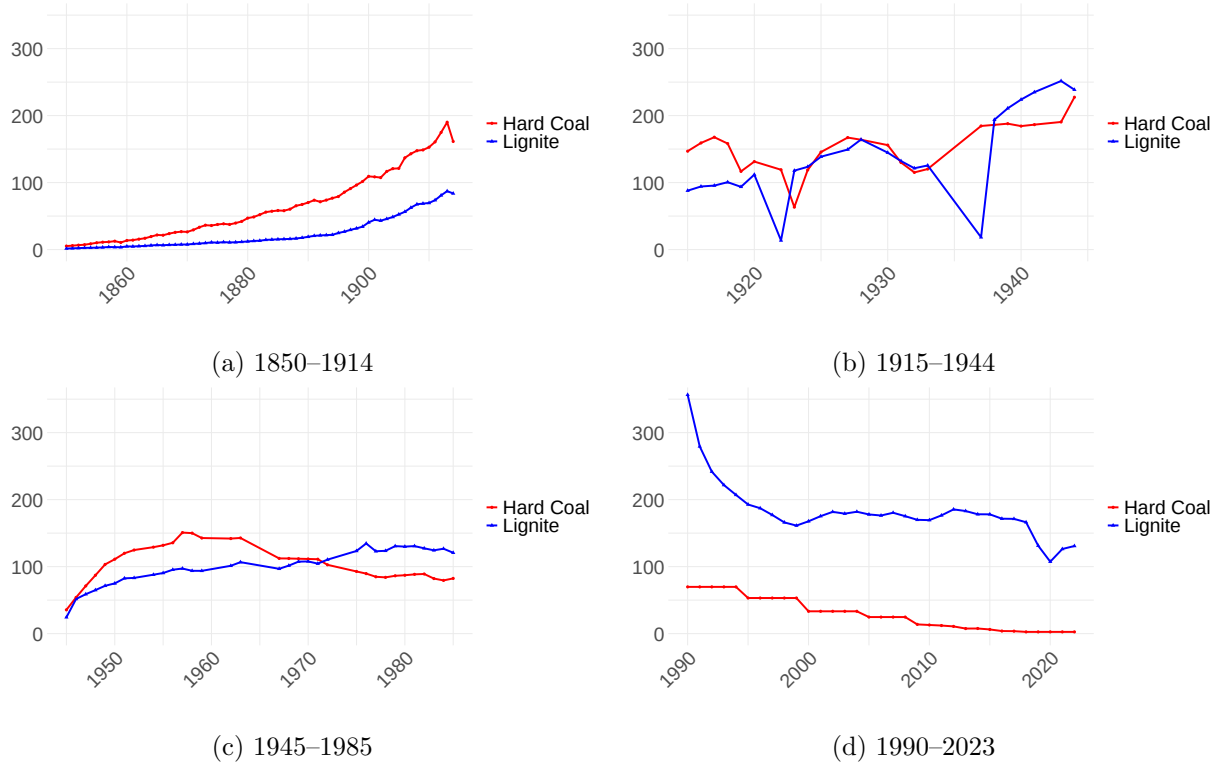


Figure A1: Hard Coal and Lignite Production over Time in Germany (1850–2023)

*Note:* This figure shows hard coal and lignite production in Germany over time, measured in million tons. Data for the period 1850–1985 are from Fischer and Fehrenbach (1989); data for 1990–2023 are from (Statista Research Department, n.d.-a, n.d.-b).

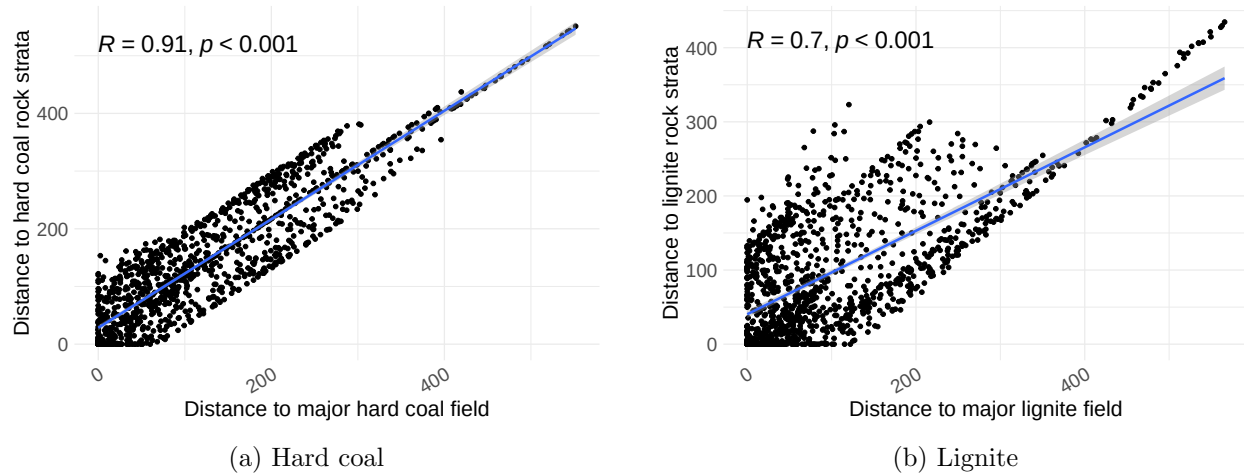


Figure A2: Correlation Between Distance to Major Coal Fields and Distance to Rock Strata in Germany

*Note:* The scatterplots show the correlation between the centroid distance of a county to the nearest large hard coal (left) or lignite (right) field and to the nearest carboniferous or lignite rock strata, respectively (in kilometer (km)). Data sources: Asch and Bellenberg, 2005; Fernihough and O'Rourke, 2021.

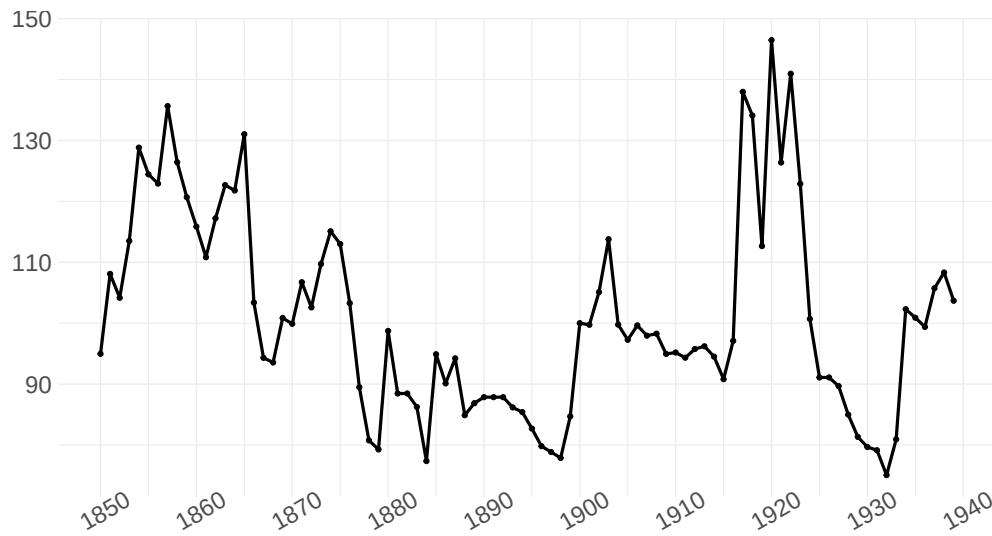


Figure A3: World Coal Prices over Time

*Note:* The figure plots annual world coal prices, 1850–1939, expressed in 1900 U.S. dollars. Source: Jacks (2019).

## A.2.2 Event-studies

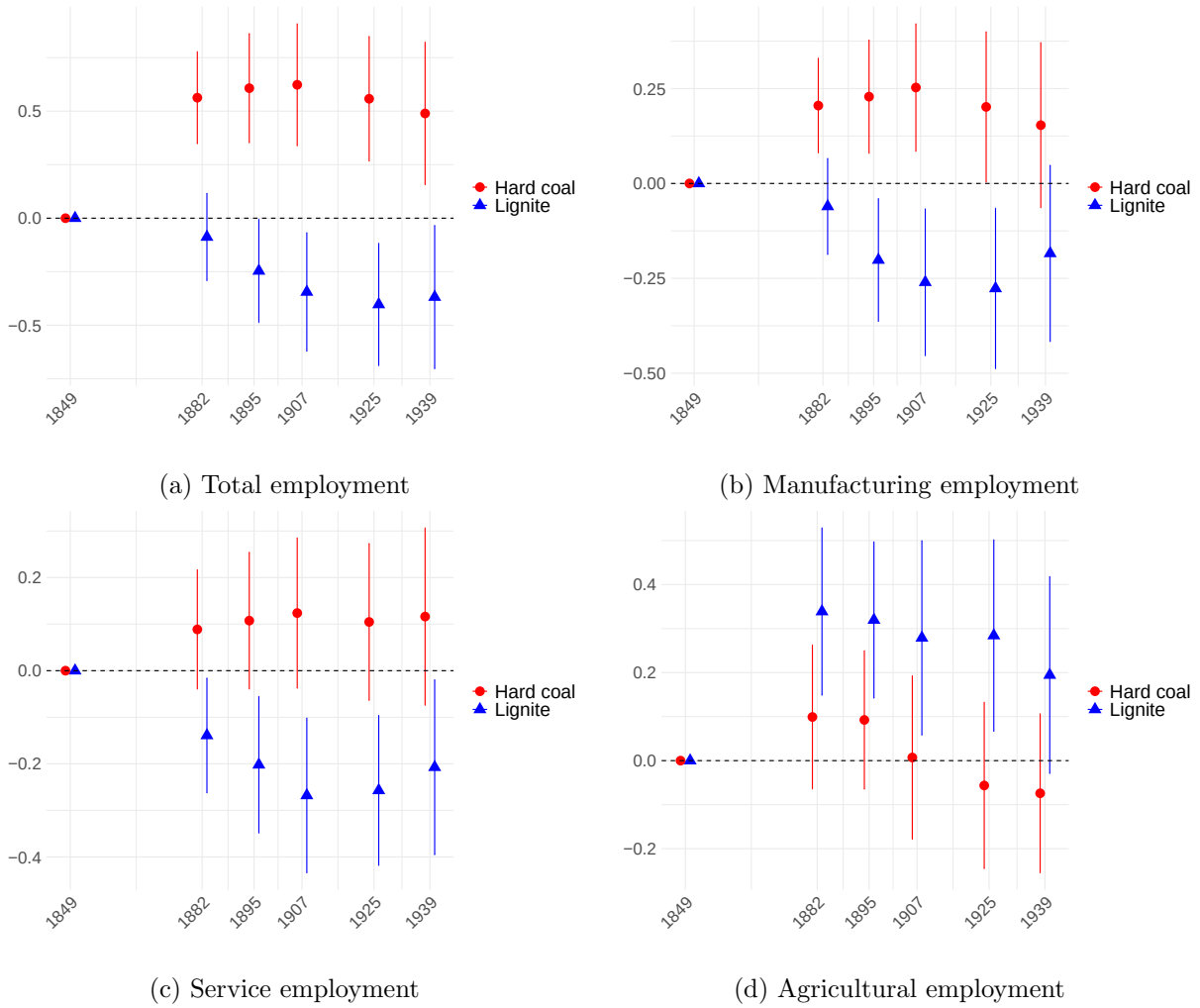


Figure A4: Event-Study: Effect of Resource Endowment on Sectoral Employment (1849–1939, Balanced Sample)

*Note:* This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1 using a balanced sample. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a standard deviation of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

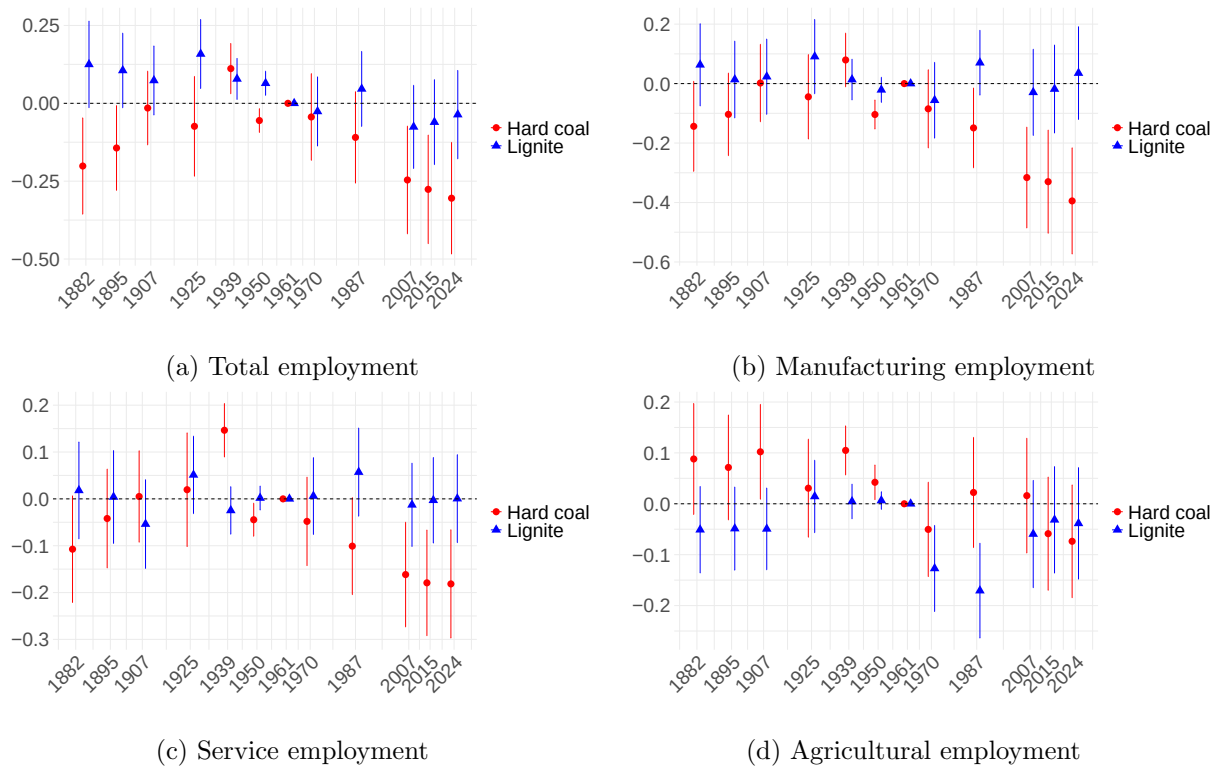


Figure A5: Event-Study: Effect of Coal Phase-Out on Sectoral Employment (1882–2024, Balanced Sample)

*Note:* This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1 using a balanced sample. Each panel reports the estimated effects of hard coal (red) and lignite (blue) exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a standard deviation of one. Hard coal and lignite indicators are interacted with year fixed effects; the omitted year is 1961. Standard errors are clustered at the county level. The analysis is restricted to West German counties.

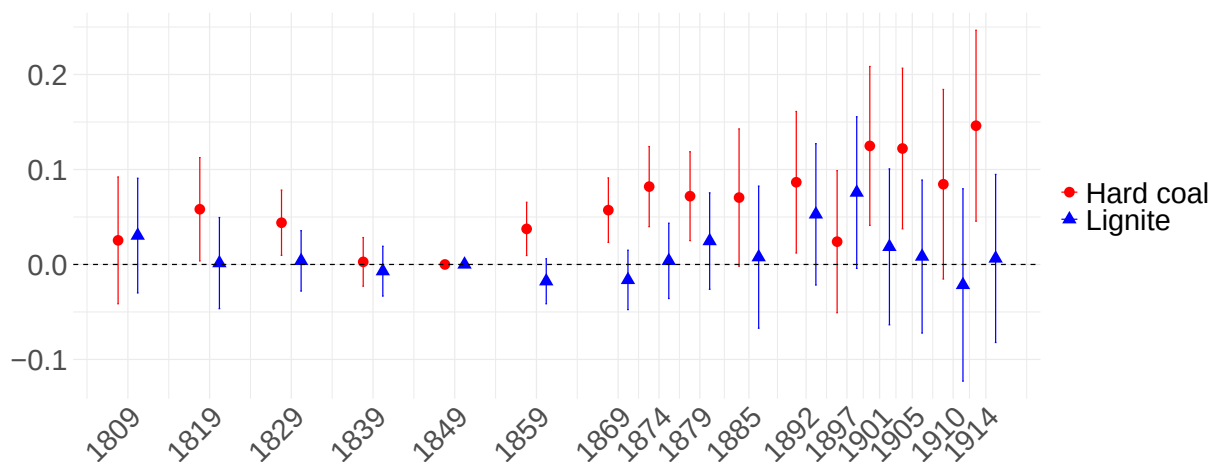


Figure A6: Event-study: Effect of Coal Endowments on Wages (1800–1914)

This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1, using unskilled wages as the outcome variable (see Section A.5 for details). The estimated effects of hard coal (red) and lignite (blue) endowment are shown relative to the omitted baseline year 1849. All regressions control for wheat prices in the nearest market, county fixed effects, and year fixed effects. Standard errors are clustered at the county level. The level of observation is the county, and the sample is restricted to Prussian counties.

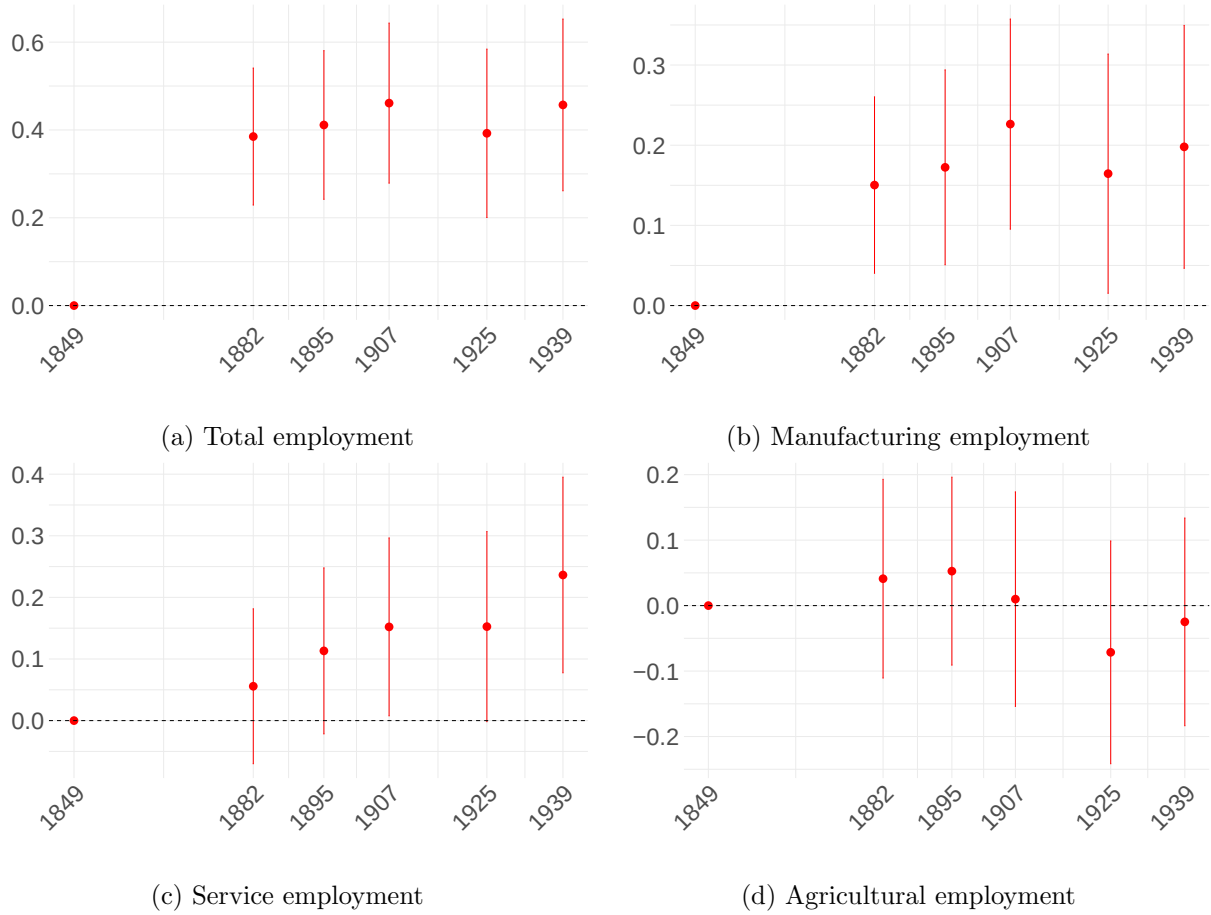


Figure A7: Event-Study: Effect of Hard Coal Endowment on Sectoral Employment (1849–1939)

*Note:* This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of hard coal exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a standard deviation of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

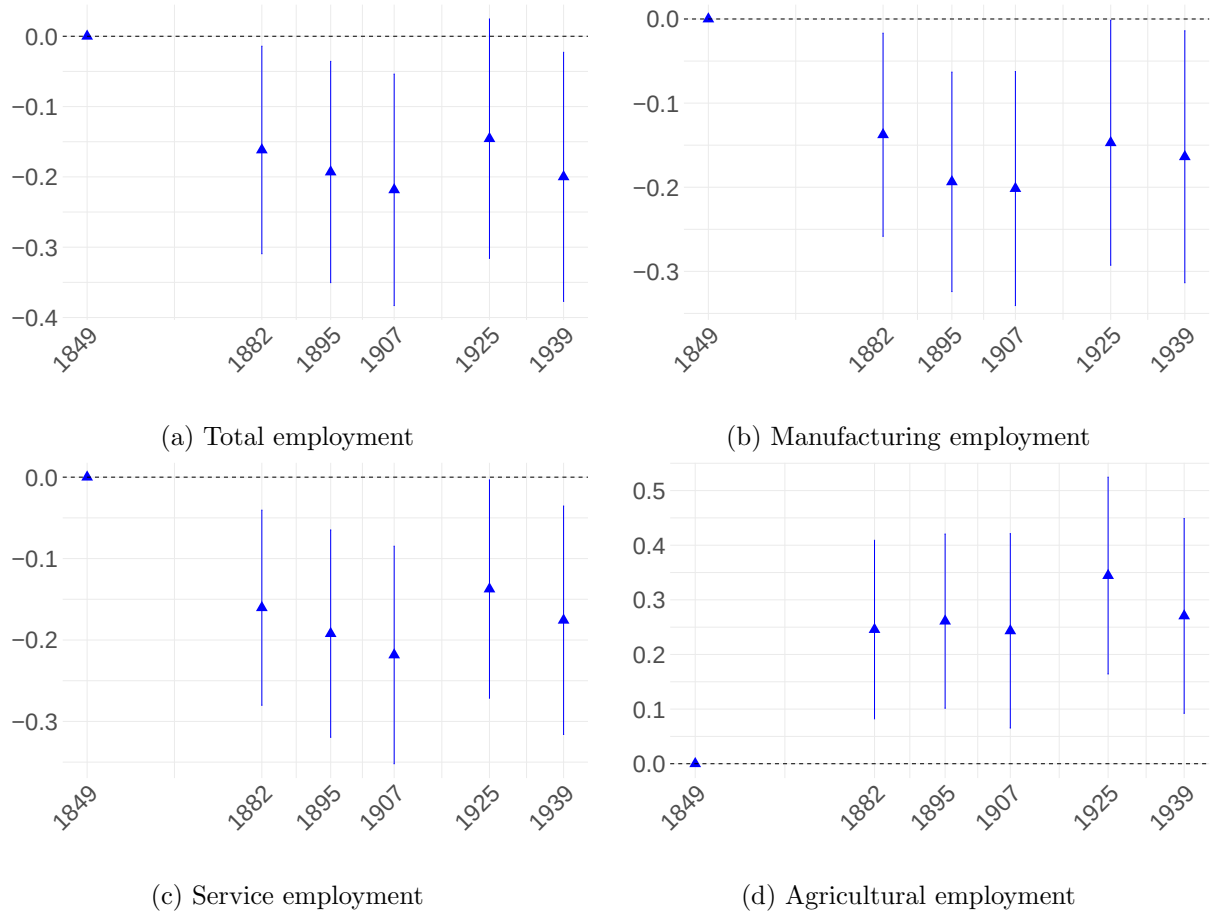


Figure A8: Event-Study: Effect of Hard Coal Endowment on Sectoral Employment (1849–1939)

*Note:* This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1. Each panel reports the estimated effects of exposure on county-level employment outcomes: (a) total employment, (b) manufacturing employment, (c) service employment, and (d) agricultural employment. All outcomes are logged and standardized to have a mean of zero and a standard deviation of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

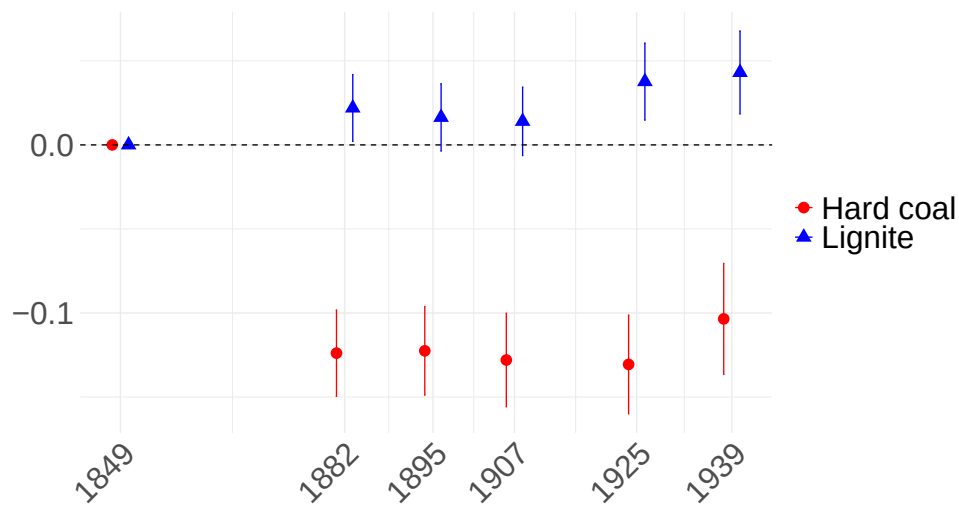


Figure A9: Event-Study: Effect of Coal Endowment on Sectoral Concentration of Manufacturing Employment (1849–1939).

*Note:* This figure plots coefficients and 95% confidence intervals from event-study regressions based on Equation 1, using the Herfindahl-Hirschman Index as the outcome variable (see Section A.3 for details). The estimated effects of hard coal (red) and lignite (blue) endowment are shown relative to the omitted baseline year 1849. The outcome is logged and standardized to have a mean of zero and a standard deviation of one. Hard coal and lignite endowment indicators are interacted with year fixed effects; the omitted year is 1849. Standard errors are clustered at the county level. The analysis is restricted to Prussian counties.

### A.2.3 Difference-in-Difference Regressions

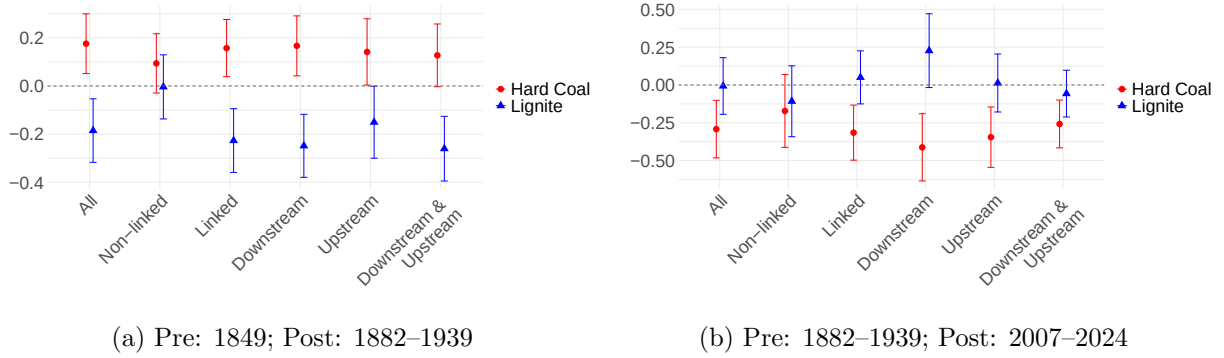


Figure A10: Difference-in-Differences regressions: Effect of Coal Endowment on Disaggregated Manufacturing Employment using German I-O table

*Note:* This figure plots coefficients and 95% confidence intervals from difference-in-differences regressions based on Equation 1, where the full set of year fixed effects is replaced by a single post-period indicator. Both panels report estimated effects of hard coal (red) and lignite (blue) endowments on county-level manufacturing employment, disaggregated by linkage to coal mining using the German IO table from 1936 (Fremdling & Tilly, 1979). All outcomes are log-transformed and standardized to have a mean of zero and a standard deviation of one. The left panel covers the extraction period (1849–1939) using counties within Prussian borders. The post-period indicator equals one for all years after 1849. The right panel shows results for the coal phase-out period (1882–2024) in West Germany, where the post-period indicator equals one for the years 2007, 2015, 2024. Standard errors are clustered at the county level.

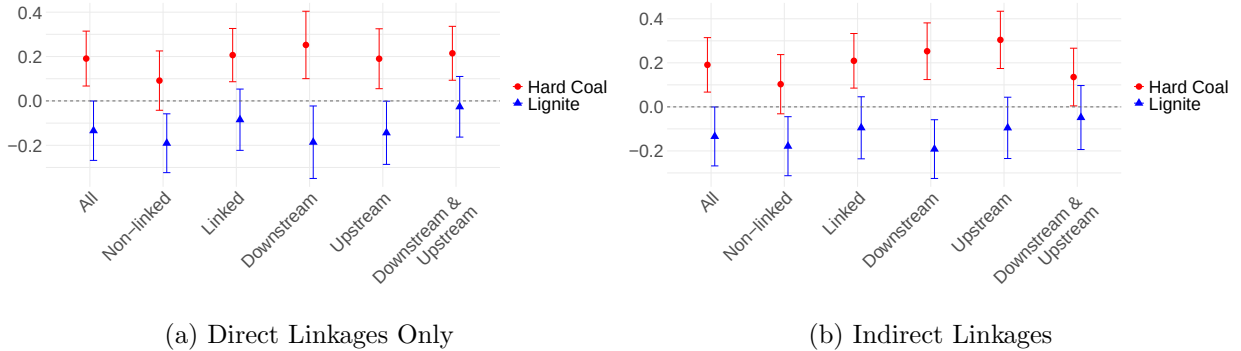


Figure A11: Difference-in-Differences Regressions: Alternative Linkage Definitions

*Note:* This figure plots coefficients and 95% confidence intervals from difference-in-differences regressions based on Equation 1, where the full set of year fixed effects is replaced by a single post-period indicator. Both panels report the estimated effects of hard coal (red) and lignite (blue) endowments on county-level manufacturing employment, disaggregated by linkage to coal mining, based on the U.S. input-output table from 1947 (U.S. Bureau of Economic Analysis, 2019). All outcomes are log-transformed and standardized to have a mean of zero and a standard deviation of one. The analysis covers the extraction period (1849–1939) and is restricted to counties within Prussian borders. The post-period indicator equals one for all years after 1849. The left panel uses direct linkages only; the right panel uses indirect linkages for both upstream and downstream relationships. Standard errors are clustered at the county level.

### A.2.4 Shift-Share Regressions

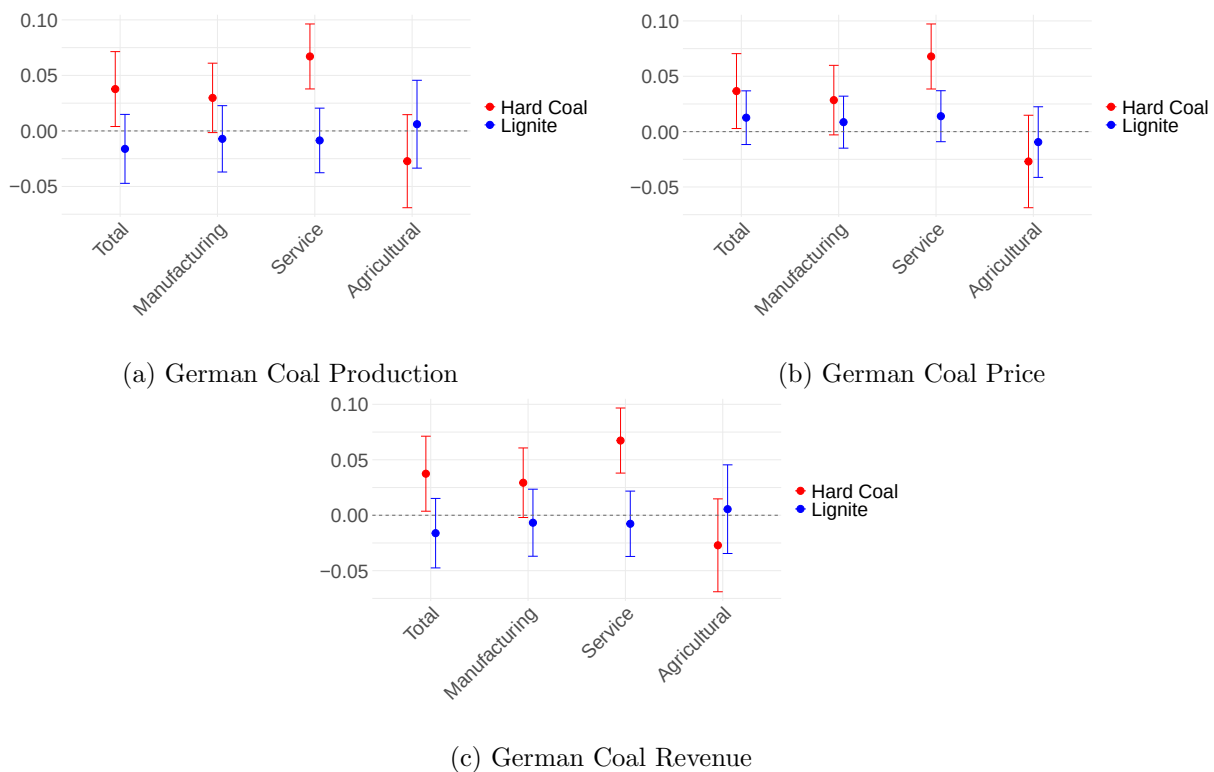
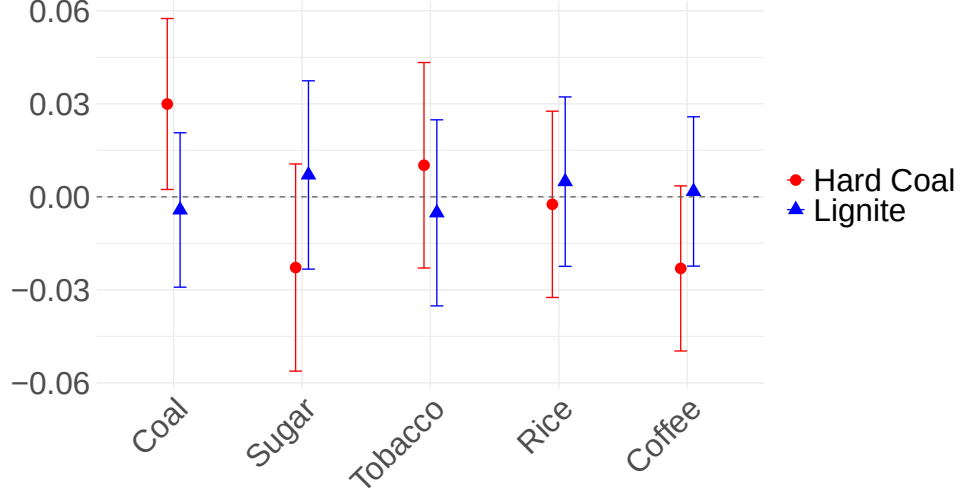


Figure A12: Reduced-Form Shift-Share Regressions: Alternative “Shift” Variables

*Note:* This figure plots coefficients and 95% confidence intervals from reduced-form shift-share regressions based on Equation 2, using alternative “shift” variables. Panel (a) uses German hard coal and lignite production; Panel (b) uses prices; and Panel (c) uses revenue (price  $\times$  quantity). Each estimate captures the differential effect of national hard coal (red) and lignite (blue) booms on counties with corresponding coal endowments, relative to non-endowed counties. The dependent variables are county-level employment in total, manufacturing, services, and agriculture. All outcomes are log-transformed and standardized to have a mean of zero and a standard deviation of one. Standard errors are clustered at the county level. The analysis covers the period 1882–1939 within the borders of the German Empire.



(a) German Coal Production

(b) Reduced-Form Shift-Share Regressions: Placebo Price Shocks

*Note:* This figure plots coefficients and 95% confidence intervals from reduced-form shift-share regressions based on Equation 2, using placebo price shocks. Each estimate captures the differential effect of global price shocks on manufacturing employment in counties with corresponding coal endowments, relative to non-endowed counties. The outcome is log-transformed and standardized to have mean zero and unit variance. Standard errors are clustered at the county level. The analysis covers the period 1882–1939 within the borders of the German Empire.

### A.3 Sectoral Concentration of Manufacturing Employment: Herfindahl–Hirschman Index

To assess whether coal mining influenced the structure of local economies, I examine its effect on industrial diversification across regions. I measure diversification using a Herfindahl–Hirschman Index (HHI) of manufacturing employment, calculated at the county-year level for each census year in which disaggregated occupational data are available.

The index is defined as the sum of squared employment shares across manufacturing industries:

$$HHI_{ct} = \sum_{i \in I} \left( \frac{\text{Employment}_{ict}}{\text{TotalEmployment}_{ct}} \right)^2$$

where  $i \in I$  indexes manufacturing industries observed in county  $c$  and year  $t$ . Higher index values reflect greater concentration (i.e., lower diversification), while lower values indicate a more diversified manufacturing structure. This measure allows me to examine whether coal extraction led to increased industrial specialization or diversification in coal-endowed regions over time, and whether these structural patterns shaped the effects of the subsequent coal phase-out.

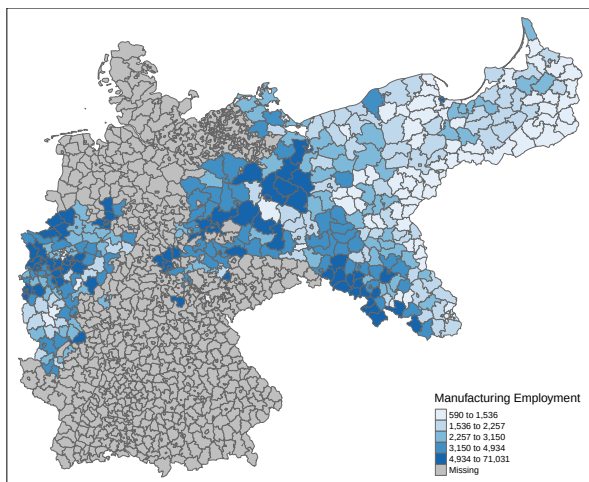
## A.4 Spatial Coverage and Harmonization of Administrative Boundaries

Administrative county borders changed repeatedly over the course of my study period due to wars, shifts in sovereignty, and territorial-administrative reforms. To ensure consistency in spatial units across time, I harmonize all data to the geographic boundaries of counties as defined in 1900. Specifically, I overlay historical shapefiles from each census year onto the 1900 county map and aggregate data by weighting historical counties according to the proportion of their area overlapping with 1900 boundaries.

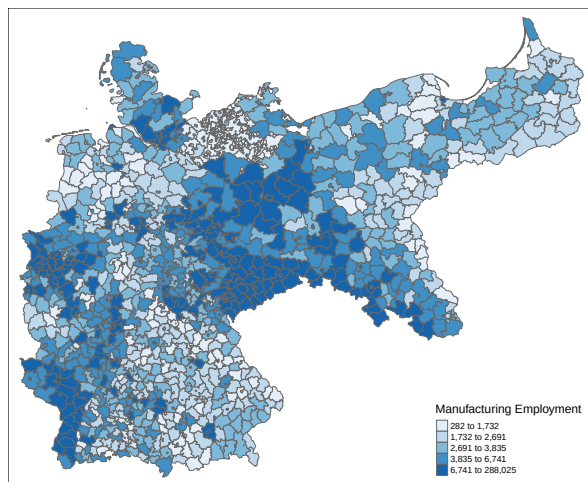
Territorial coverage also varies by year. For instance, the 1849 data include only the Kingdom of Prussia, which at the time encompassed large parts of what would later form the German Empire. In contrast, some eastern regions that were part of Germany in earlier decades, such as Silesia and East Prussia, lie outside present-day borders and are not covered in more recent census waves. To address this, I conduct separate analyses for periods with largely stable territorial coverage and provide robustness checks using balanced panels of counties observed throughout.

Figures [A14](#) and [A15](#) illustrate the geographic coverage of my dataset over time. Blue-shaded counties indicate regions for which manufacturing employment data are available in each year, while grey-shaded areas correspond to regions for which data are missing or which were not part of Germany at the time. These maps highlight how the availability of consistent county-level data varies across historical periods due to administrative boundary changes, wars, and political realignments.

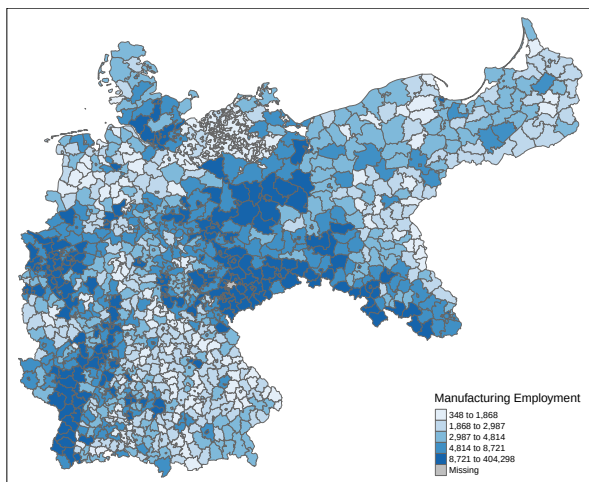
For the 1849–1939 period, I restrict the analysis to counties within Prussia, where the earliest employment data are available. For the period 1882–1939, the sample includes counties within the borders of the German Empire. For the coal phase-out period (1950–2024), I focus on counties in West Germany (Federal Republic of Germany), which offer largely stable and continuous geographic coverage over time, thereby excluding the former German Democratic Republic (GDR).



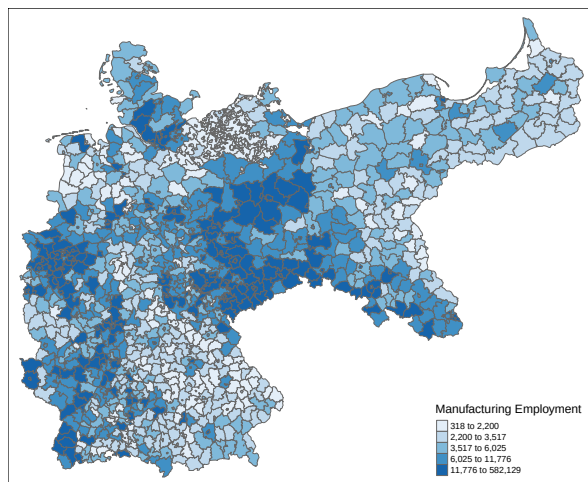
(a) 1849



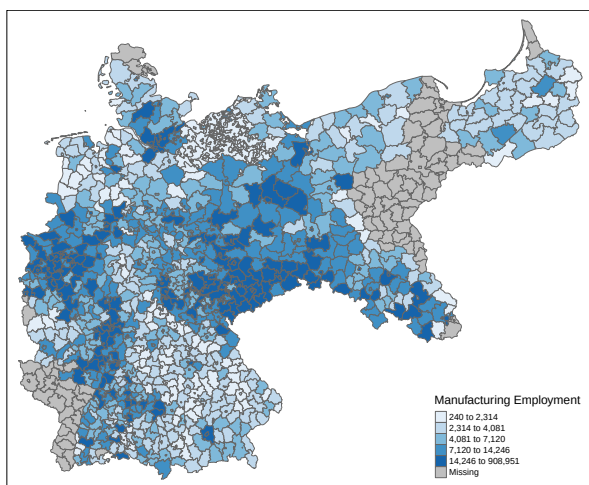
(b) 1882



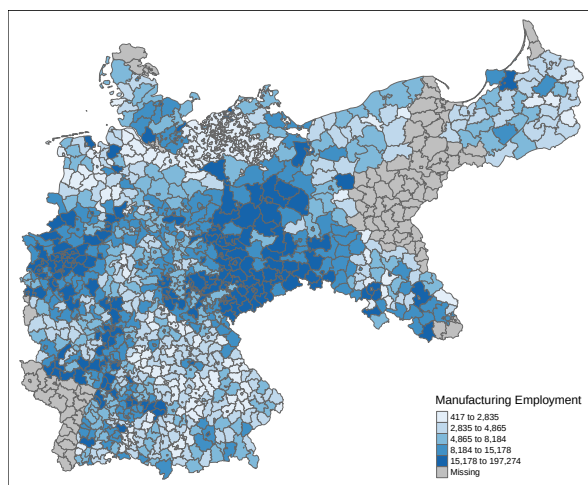
(c) 1895



(d) 1907

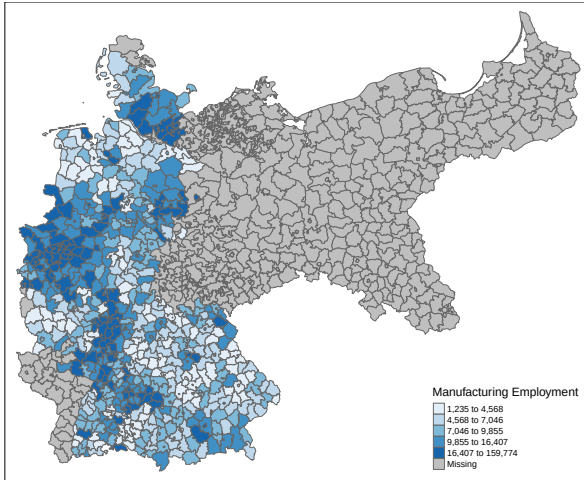


(e) 1925

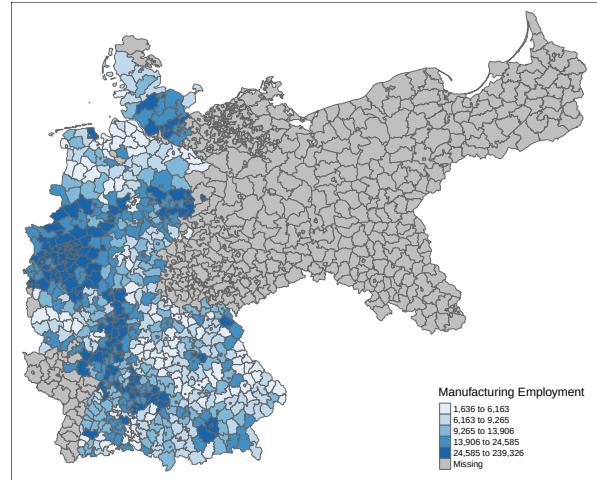


(f) 1939

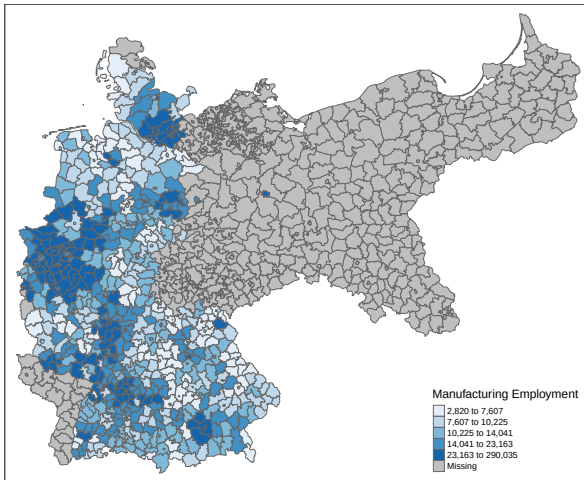
Figure A14: Manufacturing employment in German Counties (1849–1939)



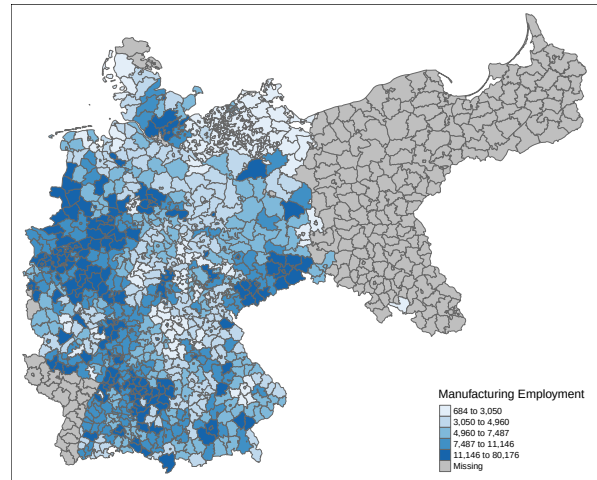
(a) 1950



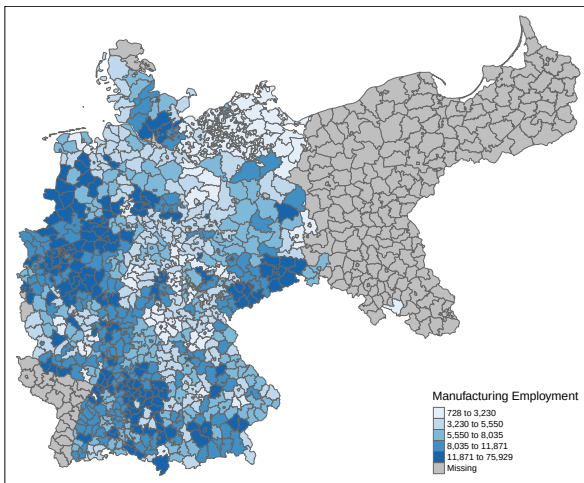
(b) 1961



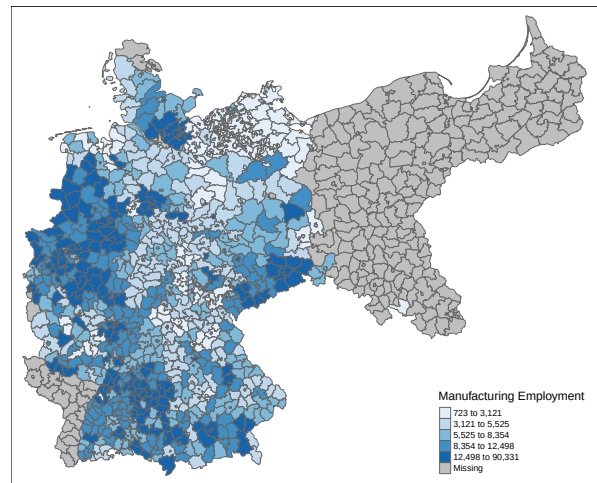
(c) 1987



(d) 2007



(e) 2015



(f) 2024

Figure A15: Manufacturing employment in German Counties (1950–2024)

## A.5 List of Variables and Data Sources

### A.5.1 Treatment Variable: Hard Coal and Lignite Endowment

**Hard Coal (Indicator):** Binary variable that takes a value of one for counties with access to hard coal resources within Germany, defined as those with rock strata formed during the Carboniferous period (approximately 295 to 320 million years ago) within 20 km of the county’s geodesic centroid; zero otherwise. Data source: Asch and Bellenberg, 2005.

**Lignite (Indicator):** Binary variable that takes a value of one for counties with access to lignite resources within Germany, defined as those with rock strata formed during the Miocene, Oligocene, Paleogene, or Pliocene period (approximately 2 to 66 million years ago) where lignite is among the two most prevalent rock types within 20 km of the county’s geodesic centroid; zero otherwise. Data source: Asch and Bellenberg, 2005.

**Hard Coal Field (Continuous):** This variable measures the distance in kilometers from a county centroid to the nearest hard coal field. The underlying geospatial data on major nineteenth-century coalfields in Europe is taken from Châtel and Dollfus (1931) and digitized by Fernihough and O’Rourke (2021). Because the dataset includes both lignite and hard coal fields, I manually classified each site based on historical sources of mine locations, retaining only hard coal fields. The distance measure is truncated at zero, that is, if a county centroid lies directly on a hard coal field, the value is set to zero.

**Lignite Field (Continuous):** This variable measures the distance in kilometers from a county centroid to the nearest lignite field. The underlying geospatial data on major nineteenth-century coalfields in Europe is taken from Châtel and Dollfus (1931) and digitized by Fernihough and O’Rourke (2021). Because the dataset includes both lignite and hard coal fields, I manually classified each site based on historical sources of mine locations, retaining only lignite fields. The distance measure is truncated at zero, that is, if a county centroid lies directly on a lignite coal field, the value is set to zero.

### A.5.2 Shift Variables

**World Coal Price** I use annual real world coal prices from Jacks (2019), expressed in 1900 U.S. dollars, for all years between 1882 and 1939.

**World Prices** I use annual real world prices for rice, tobacco, sugar, and coffee from Jacks (2019), expressed in 1900 U.S. dollars, for 1882–1939.

**National Hard Coal and Lignite Production, Prices, and Revenues** Data on national German hard-coal and lignite prices and production for 1882–1939 come from (Fischer & Fehren-

bach, 1989). Revenues for each resource are computed as the product of the average national production quantity and the corresponding average national price in a given year.

### A.5.3 Outcome Variables

**Sectoral Employment:** Employment data for the years 1882, 1895, 1907, 1925, 1933, and 1939 are drawn from the respective occupation censuses in Statistisches Reichsamt, 1884, 1897, 1910, 1929, 1936b, 1943. Employment data for 1950, 1961, 1970, and 1987 are sourced from Statistisches Bundesamt, 1953, 1964, 1972, 1989 and between 2000 and 2022 from Statistische Ämter des Bundes und der Länder, 2023.

**Unskilled Wages 1800-1914:** Average daily wage in Mark of unskilled male “seasonal fill” workers (day laborers) employed in public state forestry during the time periods 1800-1809, 1810–1819, 1820-1829, 1830-1839, 1840-1849, 1850-1859, 1860-1869, 1870-1874, and 1875-1879 as reported by the Koniglich Preussisches Statistisches Bureau (1861–1904, vol. 23). Average local daily wages in Mark for manual day laborers in a county for the year 1883 as reported by Schmitz, 1888 and for the years 1892 to 1914 as reported by Zentralblatt, 1892.

### A.5.4 Geographic Control Variables

**Temperature:** The average temperature (in degrees Celsius) in a county during the 1970–2000 period, constructed by temporally and spatially aggregating time series data on mean monthly temperature at a geospatial resolution of 30 arc seconds (i.e., grid cells of approximately 1 square kilometer), obtained from the [WorldClim \(version 2.1\) dataset](#) (Fick & Hijmans, 2017).

**Rainfall:** The average precipitation (in millimeters) in a county during the 1970–2000 period, constructed by temporally and spatially aggregating time series data on total monthly precipitation at a geospatial resolution of 30 arc seconds (i.e., grid cells of approximately 1 square kilometer), obtained from the [WorldClim \(version 2.1\) dataset](#) (Fick & Hijmans, 2017).

**Slope:** Average land slope (in degrees) within a XY km buffer around each forestry or county centroid, derived from geospatial elevation data at a resolution of 30 arc seconds (approximately 1 km × 1 km). Slope is calculated using the eight nearest neighboring cells, based on the WorldClim version 2.1 dataset (Fick & Hijmans, 2017). This measure captures local terrain steepness, with higher values indicating steeper gradients.

**Distance to Coast:** The distance in kilometers from a county centroid to the nearest coast, constructed using a map of all European coastlines provided by the European Environment Agency (EEA) (<https://www.eea.europa.eu/data-and-maps/data/eea-coastline-for-analysis-1/gis-data/europe-coastline-shapefile>)

**Distance to Navigable River:** The distance in kilometers from a county centroid to the nearest navigable river, constructed using a map of waterways (Schiffahrtsstraßen) in the German Customs Union (Zollverein) in 1850, available through the “Server for Digital Historical Maps” at the Leibniz Institute of European History at the University of Mainz (IEG, 2010).

**Caloric Suitability Index:**

The average caloric suitability index in a county, as constructed by Galor and Özak, [2016](#). This index reflects potential agricultural productivity (measured in calories) in 5’x5’ grid cells, based on the crops available for cultivation.